

Programmazione lineare intera:
metodi Branch-and-Bound

Esempio di riferimento

Un rivenditore al dettaglio acquista all'ingrosso ogni giorno 2 tipi di prodotti che, essendo deperibili, vengono rivenduti nell'arco della giornata:

- 1 cassa del prodotto 1 costa 10 euro e la vendita al dettaglio genera un profitto di 10 euro
- 1 cassa del prodotto 2 costa 19 euro e la vendita al dettaglio genera un profitto di 4 euro.

Il rivenditore ha un budget giornaliero di 190 euro.

Per portare al proprio negozio le casse acquistate, il rivenditore utilizza un furgone che può trasportare

- fino a 7 casse, se viene trasportato solo il prodotto 1 (l'ingombro di una cassa di prodotto 1 è $\frac{1}{7}$ della capacità del furgone)
- fino a 20 casse, se trasporta solo il prodotto 2 (l'ingombro di una cassa di prodotto 2 è $\frac{1}{20}$ della capacità del furgone)
- qualsiasi combinazione di casse dei 2 prodotti che non eccede la capacità del furgone.

Il rivenditore vuole determinare quante casse acquistare di ciascun prodotto per massimizzare il profitto.

Modello di PLI

$x_j \in \mathbb{N}$ n° casse di prodotto j

$$\max \omega = 10 x_1 + 4 x_2$$

$$10 x_1 + 19 x_2 \leq 190 \quad \text{vincolo di budget}$$

$$\frac{1}{7} x_1 + \frac{1}{20} x_2 \leq 1 \quad \text{vincolo di capacità}$$

$$x_1, x_2 \geq 0 \text{ e interi}$$

$$\max \omega = 10 x_1 + 4 x_2$$

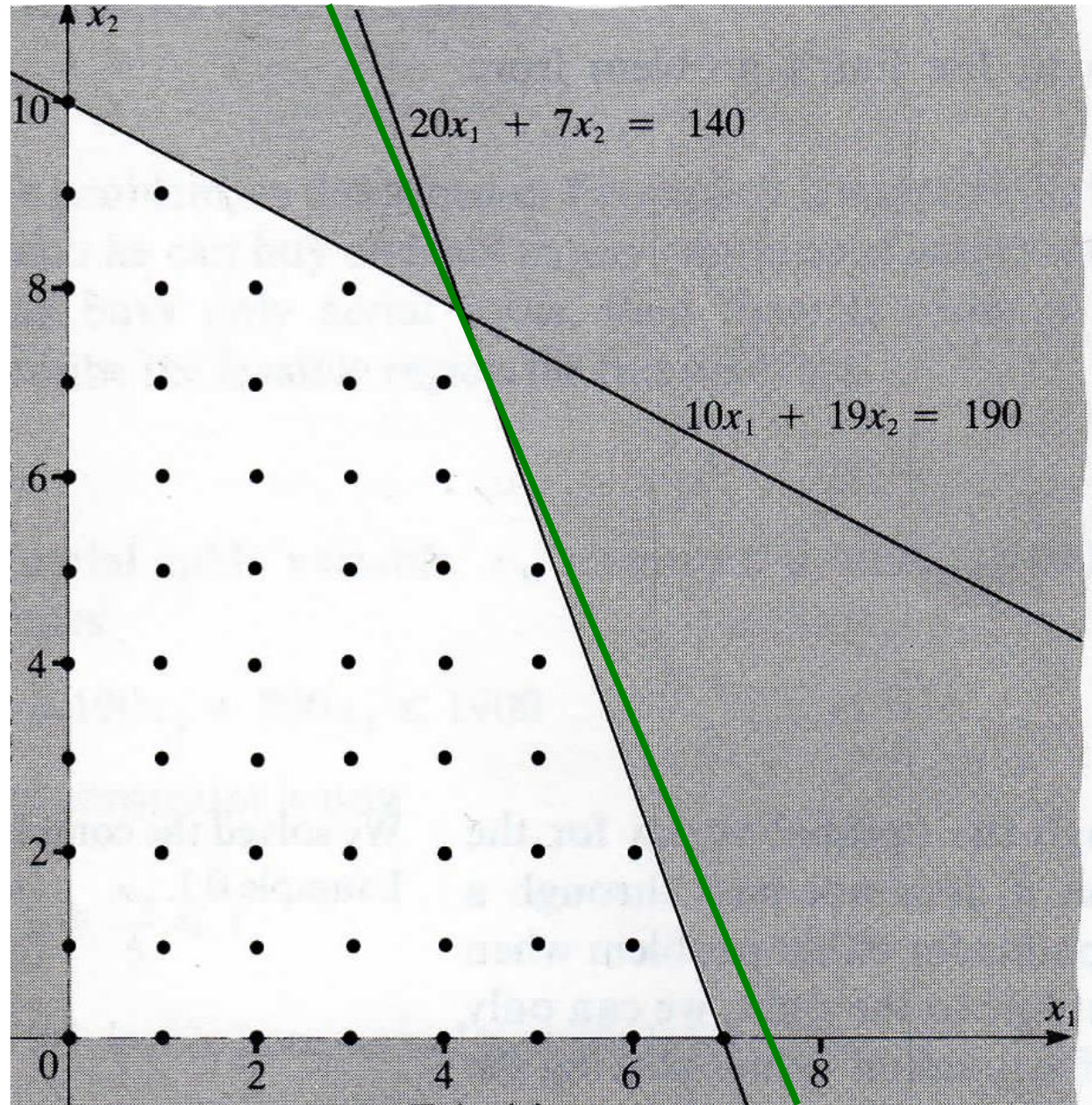
$$10 x_1 + 19 x_2 \leq 190$$

$$20 x_1 + 7 x_2 \leq 140$$

$$x_1, x_2 \geq 0 \text{ e interi}$$

← coefficienti interi

$$-\frac{20}{7} \leq -\frac{5}{2} \leq -\frac{10}{19}$$

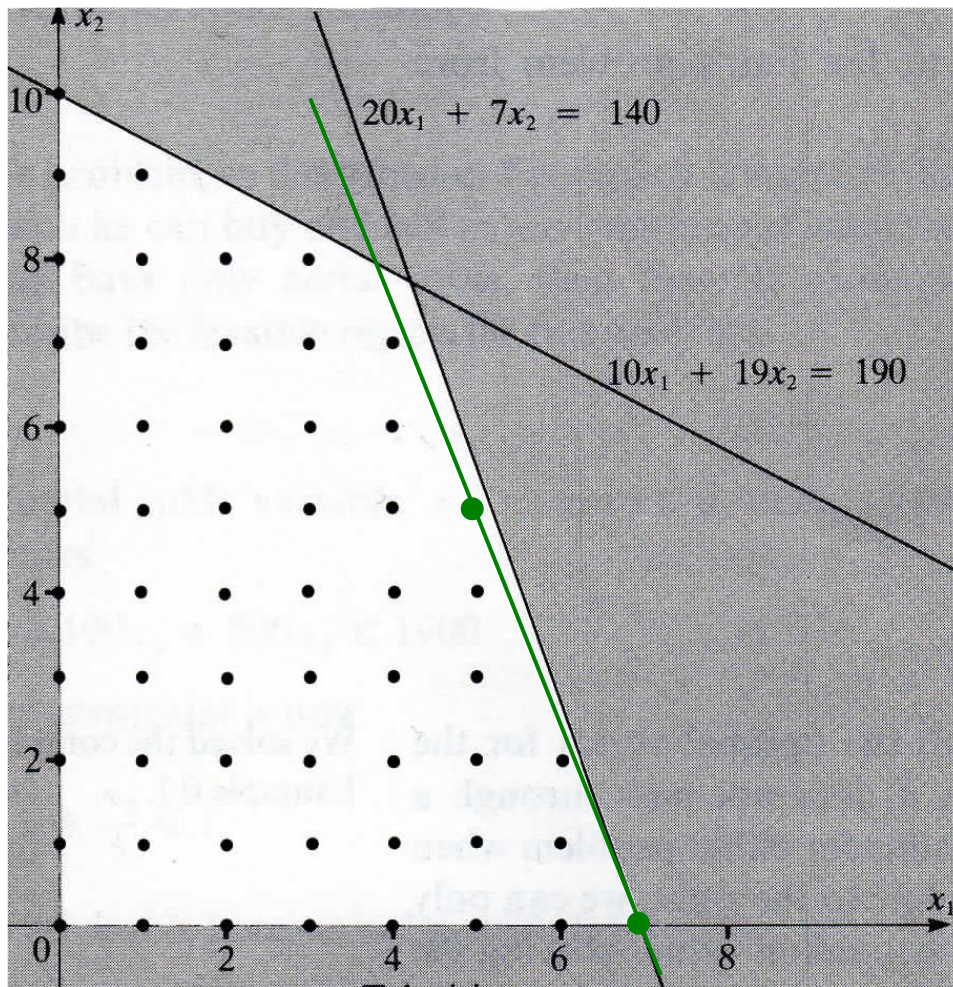


$$x_2 = -\frac{5}{2}x_1 + \frac{\omega}{4}$$

$$x_2 \leq -\frac{10}{19}x_1 + 10$$

$$x_2 \leq -\frac{20}{7}x_1 + 20$$

Con il metodo grafico si determina che il problema ha due soluzioni ottime alternative



$$\left. \begin{array}{l} x_1 = 7 \quad x_2 = 0 \\ x_1 = 5 \quad x_2 = 5 \end{array} \right\} \omega = 70$$

Soluzione del rilassamento continuo P_0

$$\max \omega = 10 x_1 + 4 x_2$$

$$10 x_1 + 19 x_2 \leq 190$$

$$20 x_1 + 7 x_2 \leq 140$$

$$x_1, x_2 \geq 0$$

P_0

x_1	x_2	s_1	s_2	φ			
10	19	1	0	0	190	$190 - 10 x_1 \geq 0$	$x_1 \leq 19$
20	7	0	1	0	140	$140 - 20 x_1 \geq 0$	$x_1 \leq 7 \quad \leftarrow$
10	4	0	0	1	0		
$\uparrow$							

$$\bar{R}_2 = \frac{1}{20} R_2$$

x_1	x_2	s_1	s_2	φ	
1	$\frac{7}{20}$	0	$\frac{1}{20}$	0	7

$$\bar{R}_1 = R_1 - 10 \bar{R}_2$$

10	19	1	0	0	190
-10	$-\frac{7}{2}$	0	$-\frac{1}{2}$	0	-70
0	$\frac{31}{2}$	1	$-\frac{1}{2}$	0	120

$$\bar{R}_0 = R_0 - 10 \bar{R}_2$$

10	4	0	0	1	0
-10	$-\frac{7}{2}$	0	$-\frac{1}{2}$	0	-70
0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	-70

x_1	x_2	s_1	s_2	φ	
0	$\frac{31}{2}$	1	$-\frac{1}{2}$	0	120
1	$\frac{7}{20}$	0	$\frac{1}{20}$	0	7
0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	-70

↑

$$120 - \frac{31}{2}x_2 \geq 0 \quad x_2 \leq \frac{240}{31} \cong 7.7 \quad \leftarrow$$

$$7 - \frac{7}{20}x_2 \geq 0 \quad x_2 \leq 20$$

$$\bar{R}_1 = \frac{2}{31}R_1$$

x_1	x_2	s_1	s_2	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	$\frac{240}{31}$

$$\bar{R}_2 = R_2 - \frac{7}{20}\bar{R}_1$$

1	$\frac{7}{20}$	0	$\frac{1}{20}$	0	7
0	$-\frac{7}{20}$	$-\frac{7}{310}$	$\frac{7}{620}$	0	$-\frac{84}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	$\frac{133}{31}$

$$\bar{R}_0 = R_0 - \frac{1}{2}\bar{R}_2$$

0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	-70
0	$-\frac{1}{2}$	$-\frac{1}{31}$	$\frac{1}{62}$	0	$-\frac{120}{31}$
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	1	$-\frac{2290}{31}$

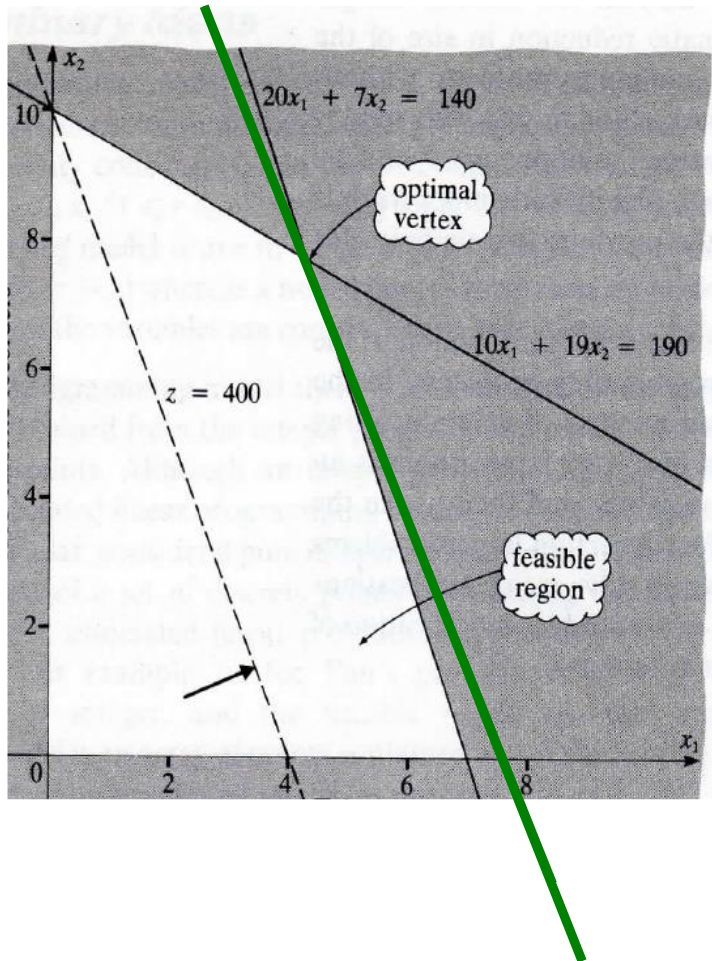
x_1	x_2	s_1	s_2	φ		
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	$\frac{240}{31}$	$= 7 + \frac{23}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	$\frac{133}{31}$	$= 4 + \frac{9}{31}$
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	1	$-\frac{2290}{31}$	$= -\left(73 + \frac{27}{31}\right)$

P_0

$$\begin{aligned} x_1 &= 4 + \frac{9}{31} \\ x_2 &= 7 + \frac{23}{31} \\ \omega &= 73 + \frac{27}{31} \end{aligned}$$

Determinazione della soluzione ottima di P_0 per via grafica

Il grafico mostra che la s.b.a.o. ottima di P_0 è il punto di intersezione delle frontiere dei due vincoli funzionali



$$\begin{cases} 10x_1 + 19x_2 = 190 \\ 20x_1 + 7x_2 = 140 \end{cases}$$

$$\begin{cases} x_2 = -\frac{10}{19}x_1 + 10 \\ x_2 = -\frac{20}{7}x_1 + 20 \end{cases}$$

$$x_1 = 4 + \frac{9}{31}$$

$$x_2 = 7 + \frac{23}{31}$$

$$\omega = 73 + \frac{27}{31}$$

Problema $P_1: P_0 \wedge x_1 \leq 4$

$$\max \omega = 10x_1 + 4x_2$$

$$10x_1 + 19x_2 \leq 190$$

$$20x_1 + 7x_2 \leq 140$$

$$x_1 \leq 4$$

$$x_1, x_2 \geq 0$$

Problema $P_2: P_0 \wedge x_1 \geq 5$

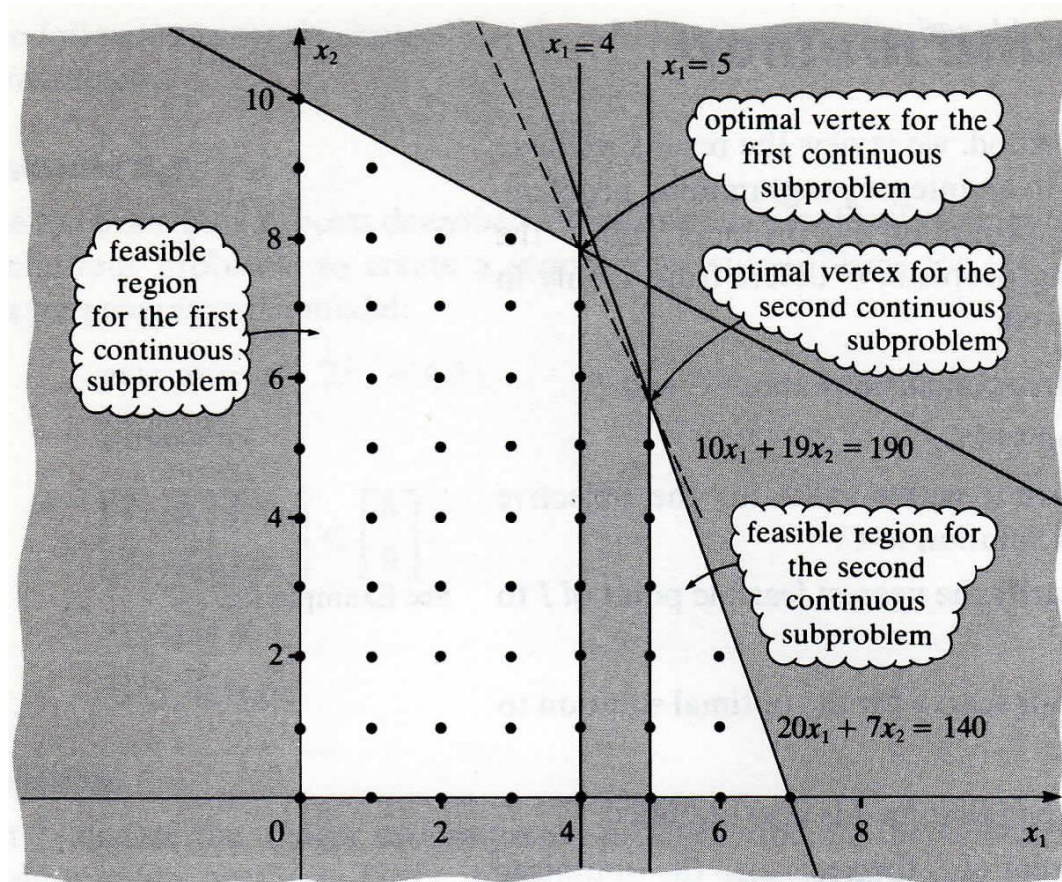
$$\max \omega = 10x_1 + 4x_2$$

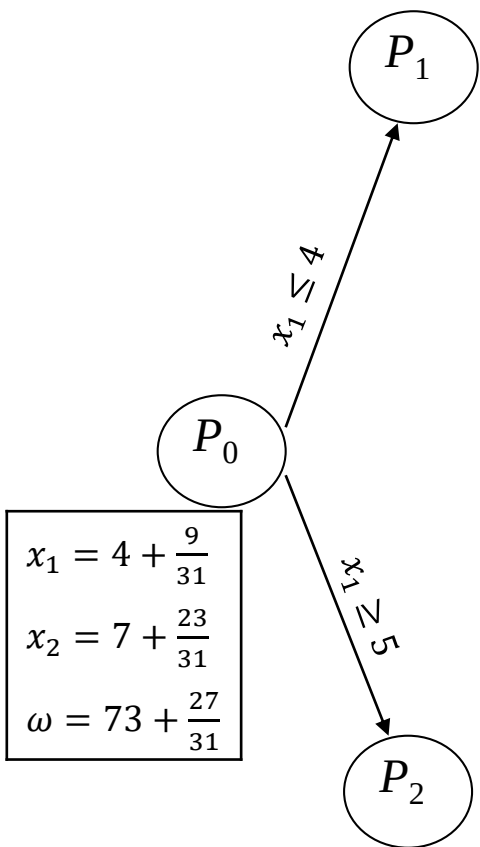
$$10x_1 + 19x_2 \leq 190$$

$$20x_1 + 7x_2 \leq 140$$

$$x_1 \geq 5$$

$$x_1, x_2 \geq 0$$





Problema P_1 : $P_0 \wedge x_1 \leq 4$

$$\max \omega = 10 x_1 + 4 x_2$$

$$10 x_1 + 19 x_2 \leq 190$$

$$20 x_1 + 7 x_2 \leq 140$$

$$x_1 \leq 4 \rightarrow x_1 + s_3 = 4$$

$$x_1, x_2 \geq 0$$

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
1	0	0	0	1	0	4
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

$$\bar{R}_3 = R_3 - R_2$$

1	0	0	0	1	0	4
-1	0	$\frac{7}{310}$	$-\frac{19}{310}$	0	0	$-\frac{133}{31}$
0	0	$\frac{7}{310}$	$-\frac{19}{310}$	1	0	$-\frac{9}{31}$

Cambio di base mediante l'algoritmo del simplesso duale

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
0	0	$\frac{7}{310}$	$-\frac{19}{310}$	1	0	$-\frac{9}{31}$ ←
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
0	0	$\frac{7}{310}$	$-\frac{19}{310}$	1	0	$-\frac{9}{31}$ ←
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

↑

$$\bar{R}_3 = -\frac{310}{19}R_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	0	$-\frac{7}{19}$	1	$-\frac{310}{19}$	0	$\frac{90}{19}$

$$\bar{R}_1 = R_1 + \frac{1}{31}\bar{R}_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
	0	0	$-\frac{7}{19 \cdot 31}$	$\frac{1}{31}$	$-\frac{10}{19}$	0	$\frac{90}{19 \cdot 31}$
	0	1	$\frac{1}{19}$	0	$-\frac{10}{19}$	0	$\frac{150}{19}$

$$\bar{R}_2 = R_2 + R_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
	0	0	$\frac{7}{310}$	$-\frac{19}{310}$	1	0	$-\frac{9}{31}$
	1	0	0	0	1	0	4

$$\bar{R}_0 = R_0 + \frac{15}{31}\bar{R}_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$
	0	0	$-\frac{7}{19} \cdot \frac{15}{31}$	$\frac{15}{31}$	$-\frac{310}{19} \cdot \frac{15}{31}$	0	$\frac{90}{19} \cdot \frac{15}{31}$
	0	0	$-\frac{4}{19}$	0	$-\frac{150}{19}$	1	$-\frac{1360}{19}$

Forma canonica della soluzione ottima di P_1

x_1	x_2	s_1	s_2	s_3	φ		
0	1	$\frac{1}{19}$	0	$-\frac{10}{19}$	0	$\frac{150}{19}$	$= 7 + \frac{17}{19}$
1	0	0	0	1	0	4	
0	0	$-\frac{7}{19}$	1	$-\frac{310}{19}$	0	$\frac{90}{19}$	
0	0	$-\frac{4}{19}$	0	$-\frac{150}{19}$	1	$-\frac{1360}{19}$	$= -\left(71 + \frac{11}{19}\right)$

$$\begin{aligned} x_1 &= 4 \\ x_2 &= 7 + \frac{17}{19} \\ \omega &= 71 + \frac{11}{19} \end{aligned}$$

P_1

$$x_1 \leq 4$$

P_0

$$x_1 \geq 5$$

P_2

$$\begin{aligned} x_1 &= 4 + \frac{9}{31} \\ x_2 &= 7 + \frac{23}{31} \\ \omega &= 73 + \frac{27}{31} \end{aligned}$$

Problema P_2 : $P_0 \wedge x_1 \geq 5$

$$\max \omega = 10 x_1 + 4 x_2$$

$$10 x_1 + 19 x_2 \leq 190$$

$$20 x_1 + 7 x_2 \leq 140$$

$$x_1 \geq 5 \rightarrow -x_1 + s_3 = -5$$

$$x_1, x_2 \geq 0$$

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
-1	0	0	0	1	0	-5
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

$$\bar{R}_3 = R_3 + R_2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -5$$

1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$

0	0	$-\frac{7}{310}$	$\frac{19}{310}$	1	0	$-\frac{22}{31}$

In alternativa:

$$\rightarrow x_1 - s_3 = 5$$

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
1	0	0	0	-1	0	5
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

$$\bar{R}_3 = R_3 - R_2$$

1	0	0	0	-1	0	5
-1	0	$\frac{7}{310}$	$-\frac{19}{310}$	0	0	$-\frac{133}{31}$
0	0	$\frac{7}{310}$	$-\frac{19}{310}$	-1	0	$\frac{22}{31}$

$$\overline{\bar{R}_3} = -\bar{R}_3$$

x_1	x_2	s_1	s_2	s_3	φ	
0	0	$-\frac{7}{310}$	$\frac{19}{310}$	1	0	$-\frac{22}{31}$

Cambio di base mediante l'algoritmo del simplesso duale

x_1	x_2	s_1	s_2	s_3	φ	
0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
0	0	$-\frac{7}{310}$	$\frac{19}{310}$	1	0	$-\frac{22}{31}$ ←
0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$

↑

$$\bar{R}_3 = -\frac{310}{7}R_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	0	1	$-\frac{19}{7}$	$-\frac{310}{7}$	0	$\frac{220}{7}$

$$\bar{R}_1 = R_1 - \frac{2}{31}\bar{R}_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	1	$\frac{2}{31}$	$-\frac{1}{31}$	0	0	$\frac{240}{31}$
	0	0	$-\frac{2}{31}$	$\frac{2}{31}\frac{19}{7}$	$\frac{20}{7}$	0	$-\frac{2}{31}\frac{220}{7}$
	0	1	0	$\frac{1}{7}$	$\frac{20}{7}$	0	$\frac{40}{7}$

$$\bar{R}_2 = R_2 - R_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	1	0	$-\frac{7}{310}$	$\frac{19}{310}$	0	0	$\frac{133}{31}$
	0	0	$\frac{7}{310}$	$-\frac{19}{310}$	1	0	$-\frac{22}{31}$
	1	0	0	0	-1	0	5

$$\bar{R}_0 = R_0 + \frac{1}{31}\bar{R}_3$$

	x_1	x_2	s_1	s_2	s_3	φ	
	0	0	$-\frac{1}{31}$	$-\frac{15}{31}$	0	1	$-\frac{2290}{31}$
	0	0	$\frac{1}{31}$	$-\frac{19}{7\cdot 31}$	$-\frac{10}{7}$	0	$\frac{220}{7\cdot 31}$
	0	0	0	$-\frac{4}{7}$	$-\frac{10}{7}$	1	$-\frac{510}{7}$

Forma canonica della soluzione ottima di P_2

x_1	x_2	s_1	s_2	s_3	φ		
0	1	0	$\frac{1}{7}$	$\frac{20}{7}$	0	$\frac{40}{7}$	$= 5 + \frac{5}{7}$
1	0	0	0	-1	0	5	
0	0	1	$-\frac{19}{7}$	$-\frac{310}{7}$	0	$\frac{220}{7}$	
0	0	0	$-\frac{4}{7}$	$-\frac{10}{7}$	1	$-\frac{510}{7}$	$= -\left(72 + \frac{6}{7}\right)$

$$\begin{aligned}x_1 &= 4 \\x_2 &= 7 + \frac{17}{19} \\\omega &= 71 + \frac{11}{19}\end{aligned}$$

$$P_1$$

$$x_1 \leq 4$$

$$P_0$$

$$x_1 \geq 5$$

$$P_2$$

$$\begin{aligned}x_1 &= 4 + \frac{9}{31} \\x_2 &= 7 + \frac{23}{31} \\\omega &= 73 + \frac{27}{31}\end{aligned}$$

$$\begin{aligned}x_1 &= 5 \\x_2 &= 5 + \frac{5}{7} \\\omega &= 72 + \frac{6}{7}\end{aligned}$$

Problema P_3 : $P_0 \wedge x_1 \geq 5 \wedge x_2 \leq 5$

$$\max \omega = 10x_1 + 4x_2$$

$$10x_1 + 19x_2 \leq 190$$

$$20x_1 + 7x_2 \leq 140$$

$$x_1 \geq 5$$

$$x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

Problema P_4 : $P_0 \wedge x_1 \geq 5 \wedge x_2 \geq 6$

$$\max \omega = 10x_1 + 4x_2$$

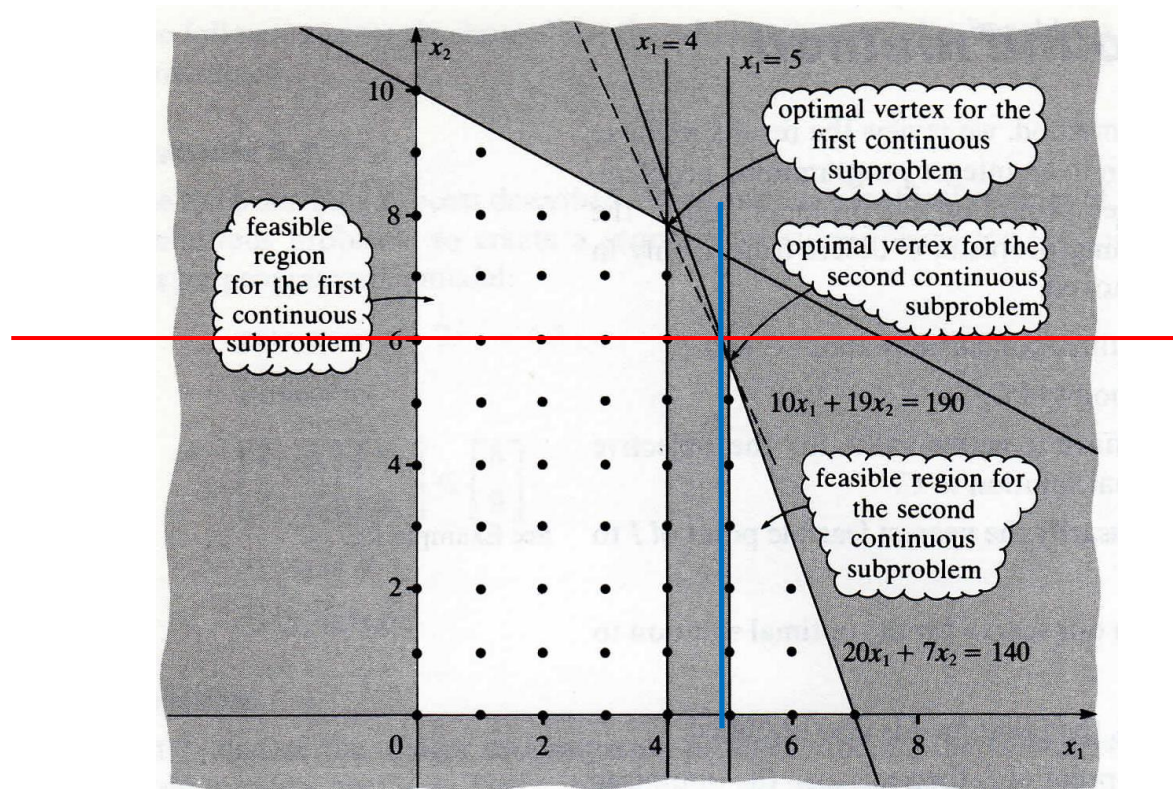
$$10x_1 + 19x_2 \leq 190$$

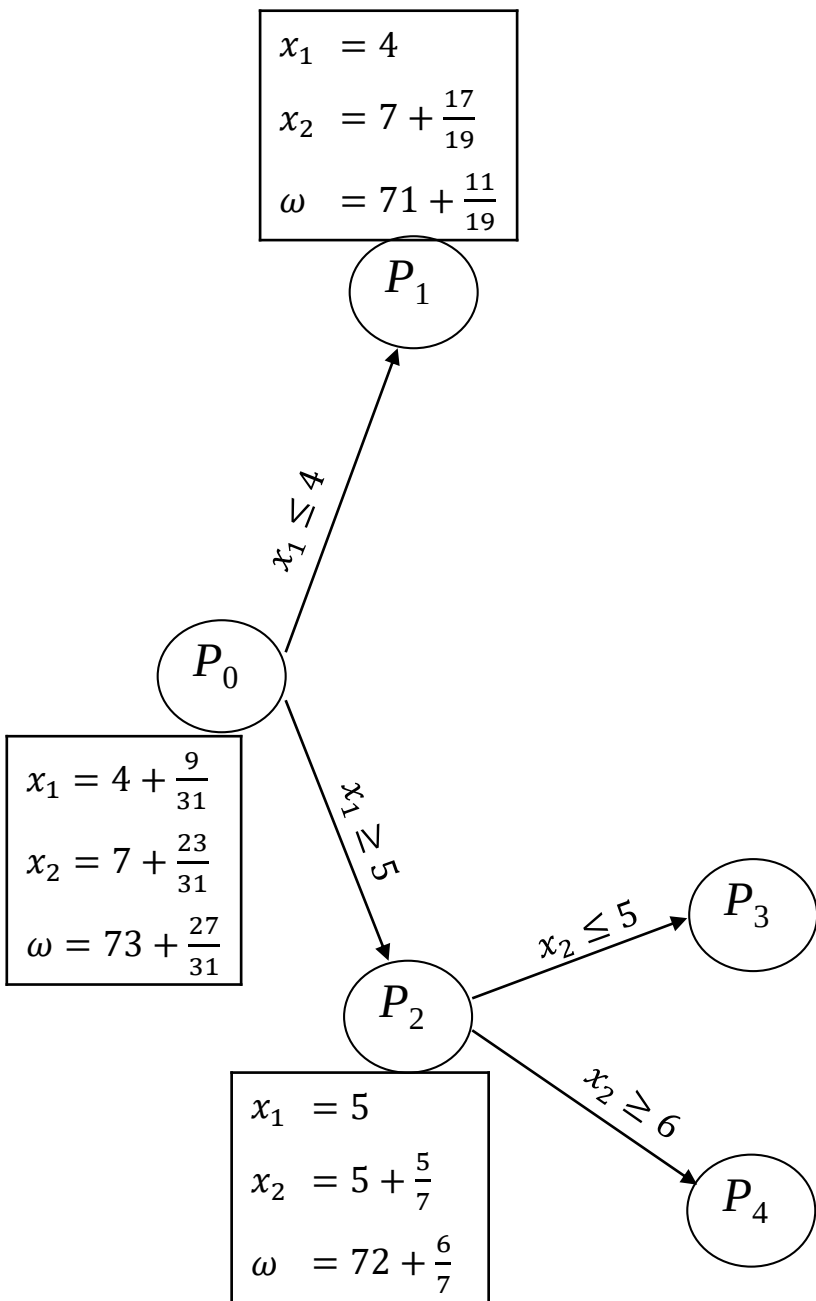
$$20x_1 + 7x_2 \leq 140$$

$$x_1 \geq 5$$

$$x_2 \geq 6$$

$$x_1, x_2 \geq 0$$





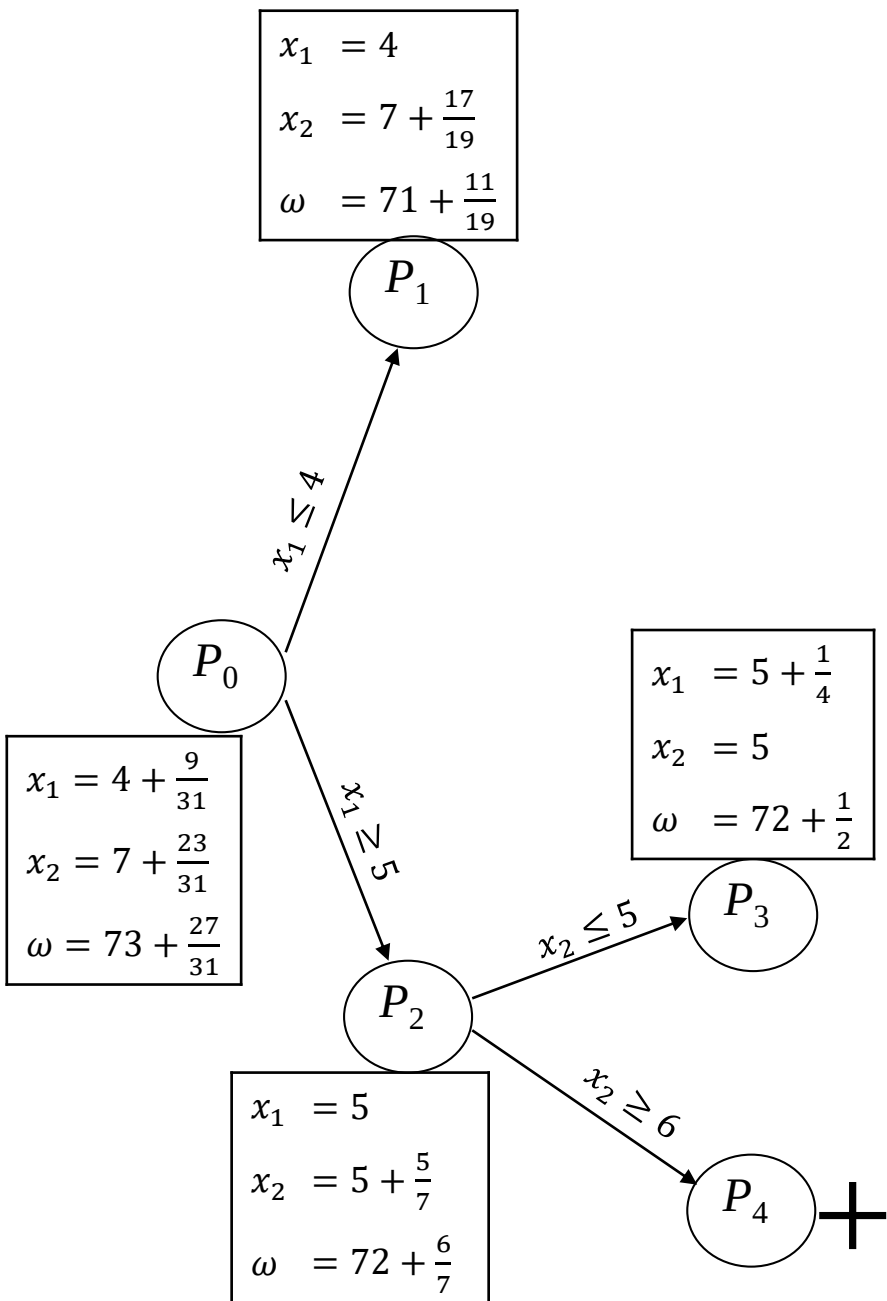
Problema P_3 : $P_0 \wedge x_1 \geq 5 \wedge x_2 \leq 5$

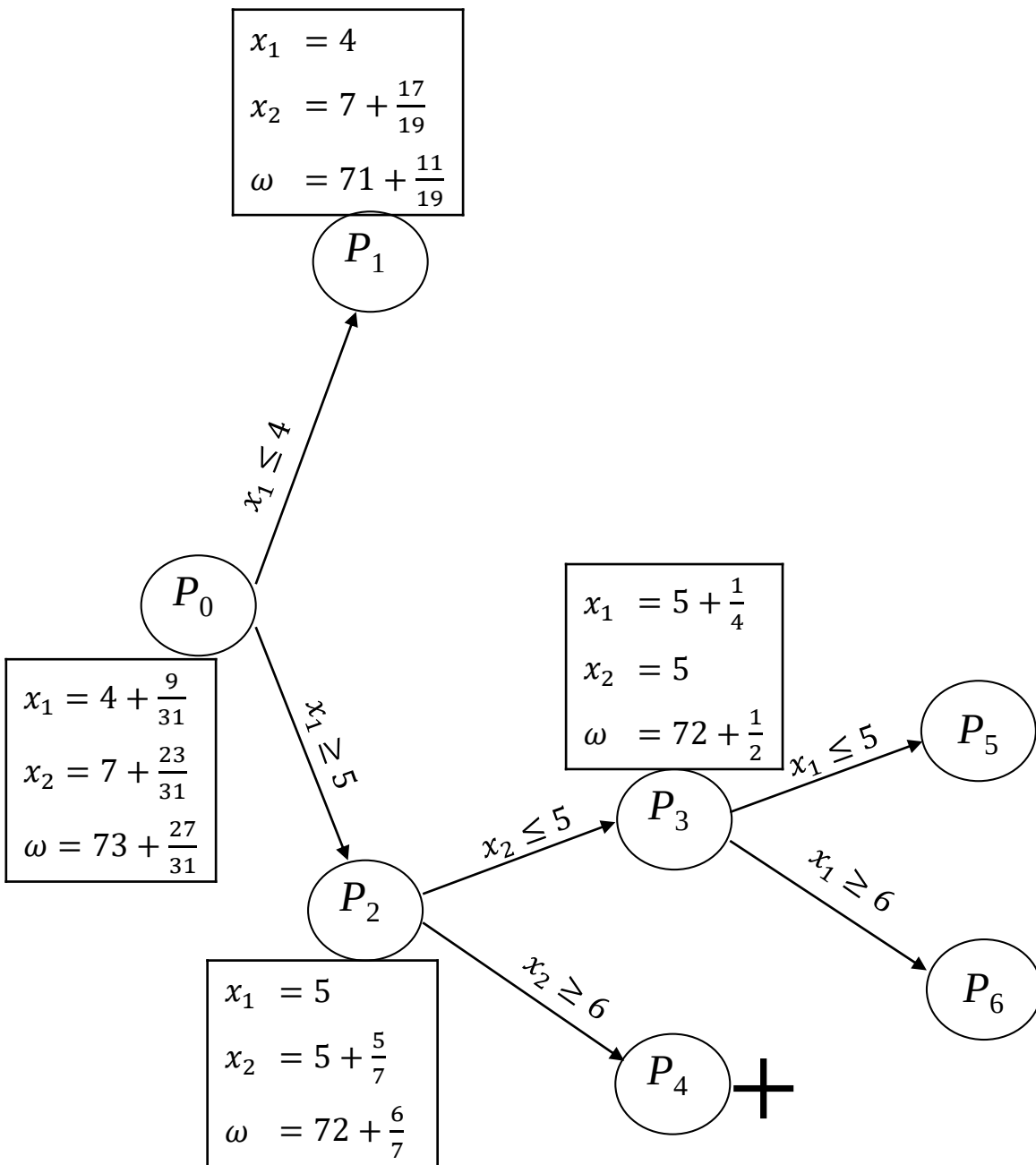
x_1	x_2	s_1	s_2	s_3	s_4	φ	
0	1	0	$\frac{1}{7}$	$\frac{20}{7}$	0	0	$\frac{40}{7}$
1	0	0	0	-1	0	0	5
0	0	1	$-\frac{19}{7}$	$-\frac{310}{7}$	0	0	$\frac{220}{7}$
0	1	0	0	0	1	0	5
0	0	0	$-\frac{4}{7}$	$-\frac{10}{7}$	0	1	$-\frac{510}{7}$

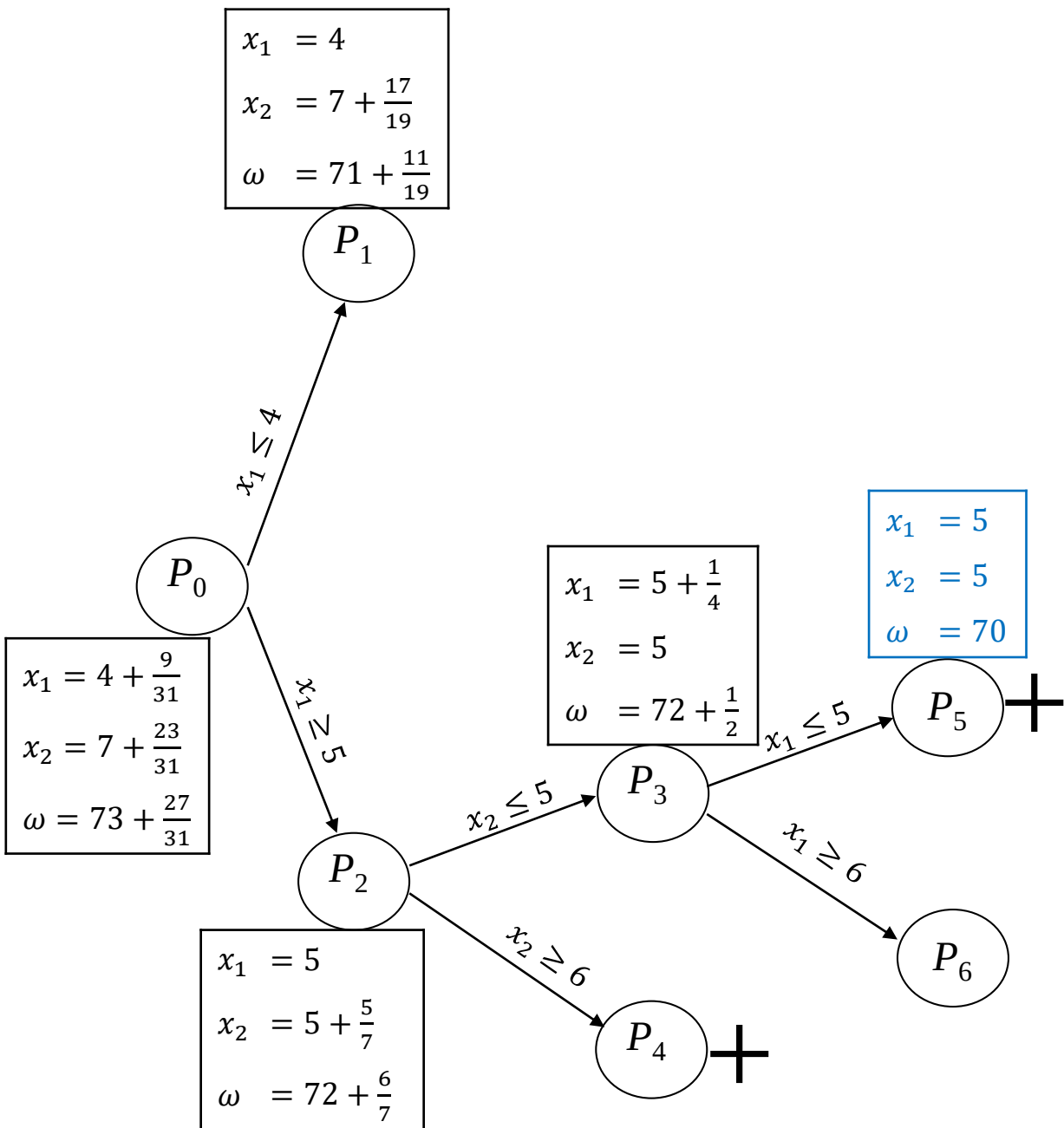
$$\bar{R}_4 = R_4 - R_1$$

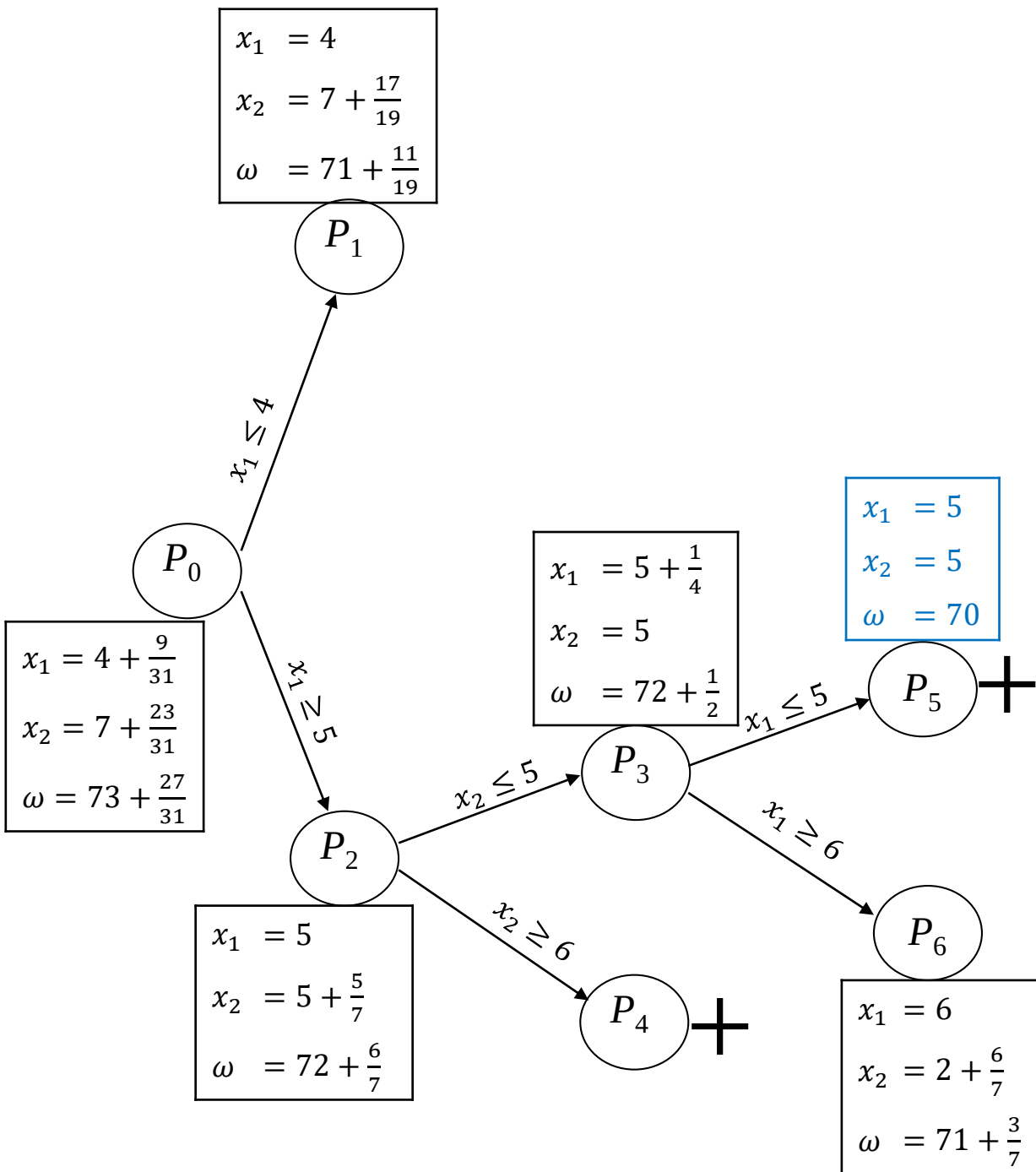
x_1	x_2	s_1	s_2	s_3	s_4	φ	
0	1	0	0	0	1	0	5
0	-1	0	$-\frac{1}{7}$	$-\frac{20}{7}$	0	0	$-\frac{40}{7}$
0	0	0	$-\frac{1}{7}$	$-\frac{20}{7}$	1	0	$-\frac{5}{7}$

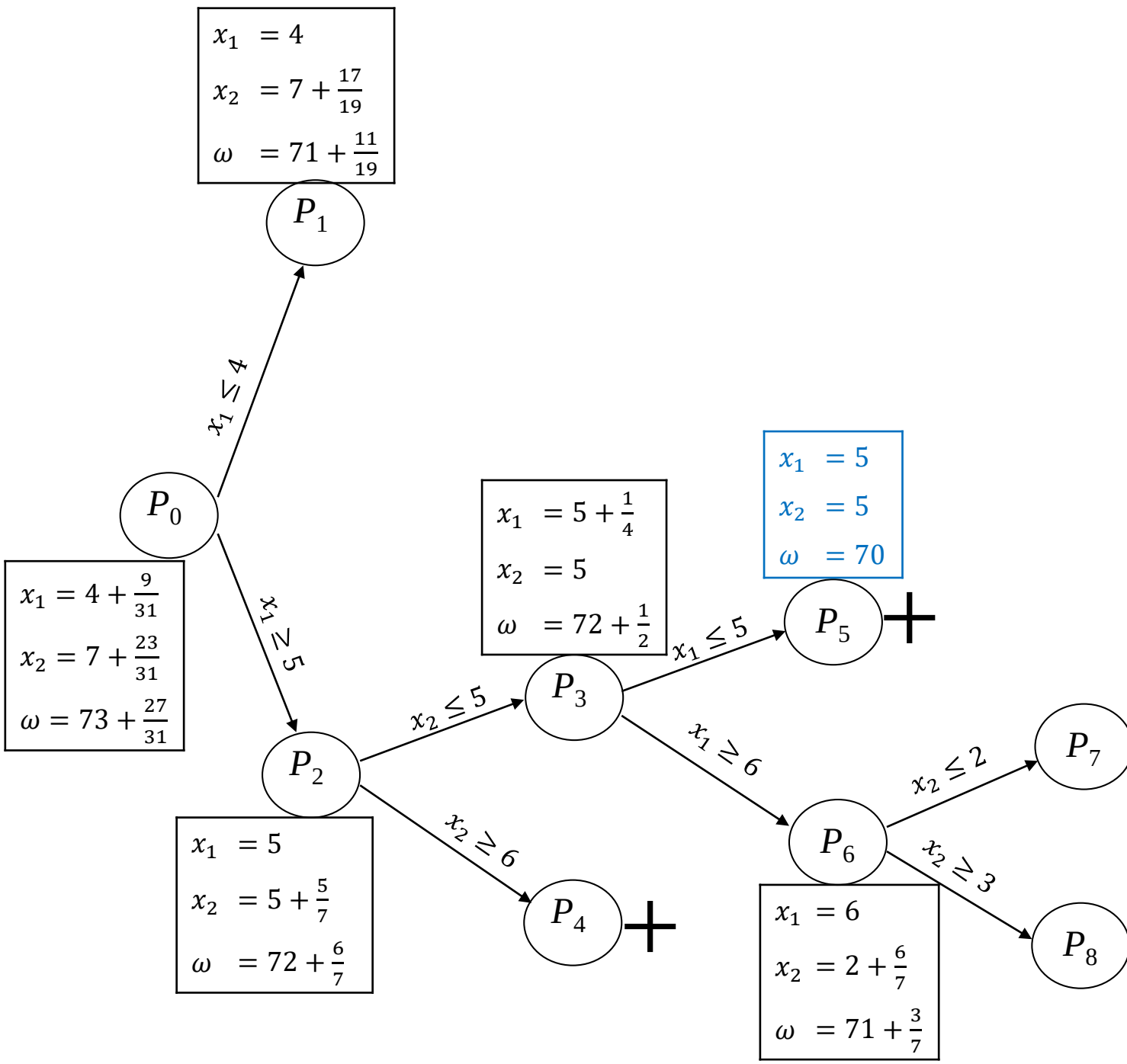
x_1	x_2	s_1	s_2	s_3	s_4	φ	
0	1	0	$\frac{1}{7}$	$\frac{20}{7}$	0	0	$\frac{40}{7}$
1	0	0	0	-1	0	0	5
0	0	1	$-\frac{19}{7}$	$-\frac{310}{7}$	0	0	$\frac{220}{7}$
0	0	0	$-\frac{1}{7}$	$-\frac{20}{7}$	1	0	$-\frac{5}{7}$
0	0	0	$-\frac{4}{7}$	$-\frac{10}{7}$	0	1	$-\frac{510}{7}$

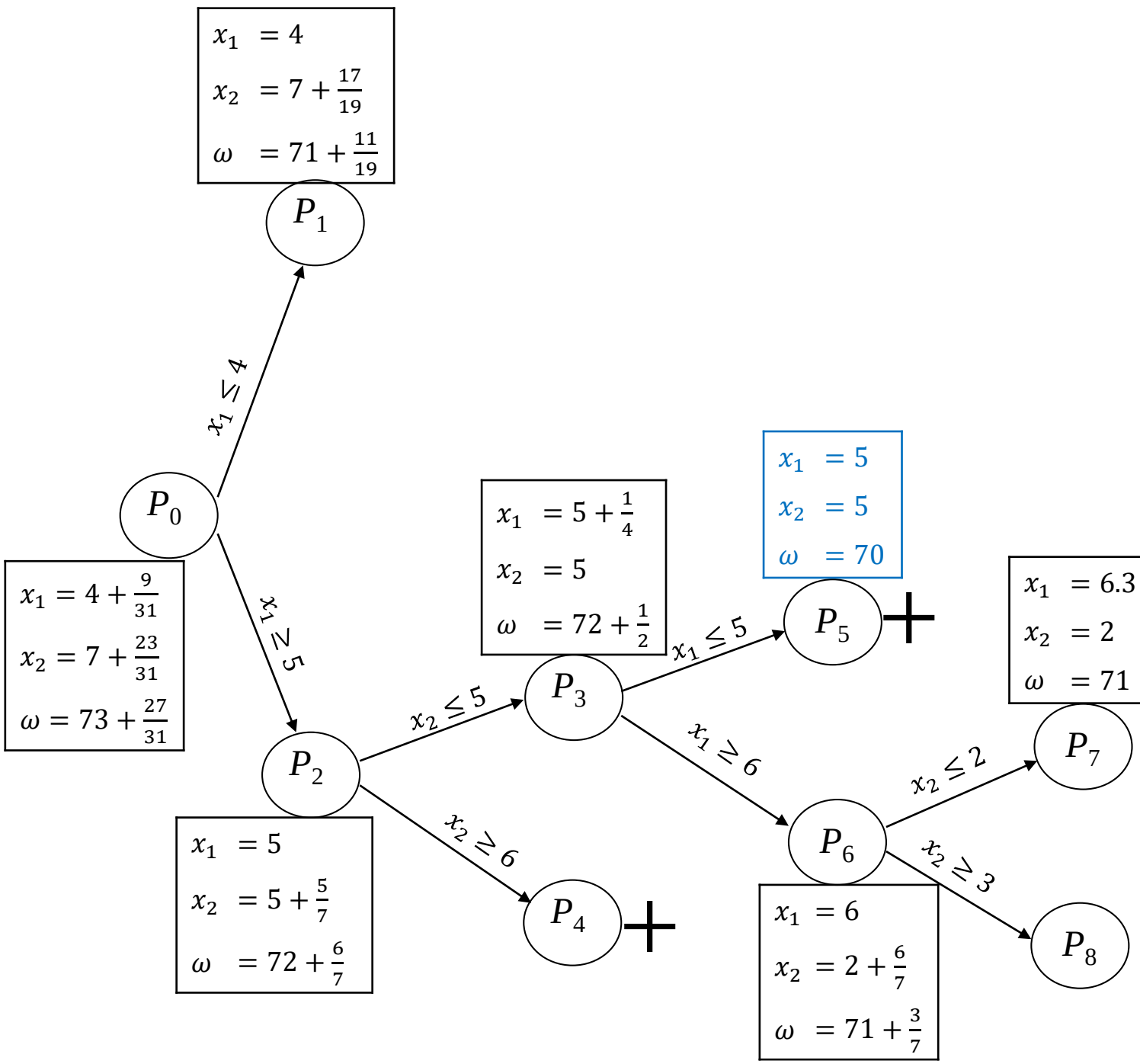


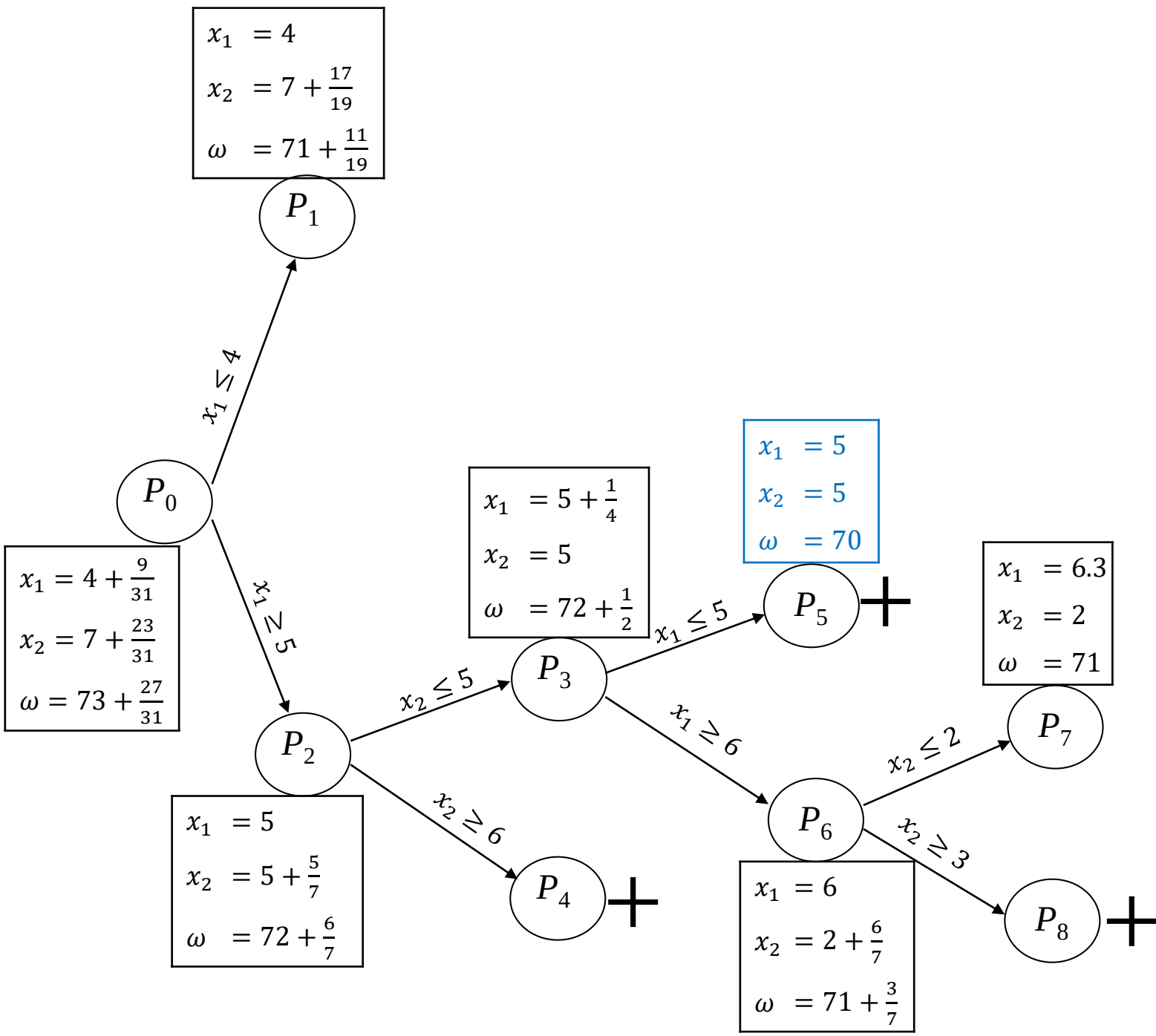


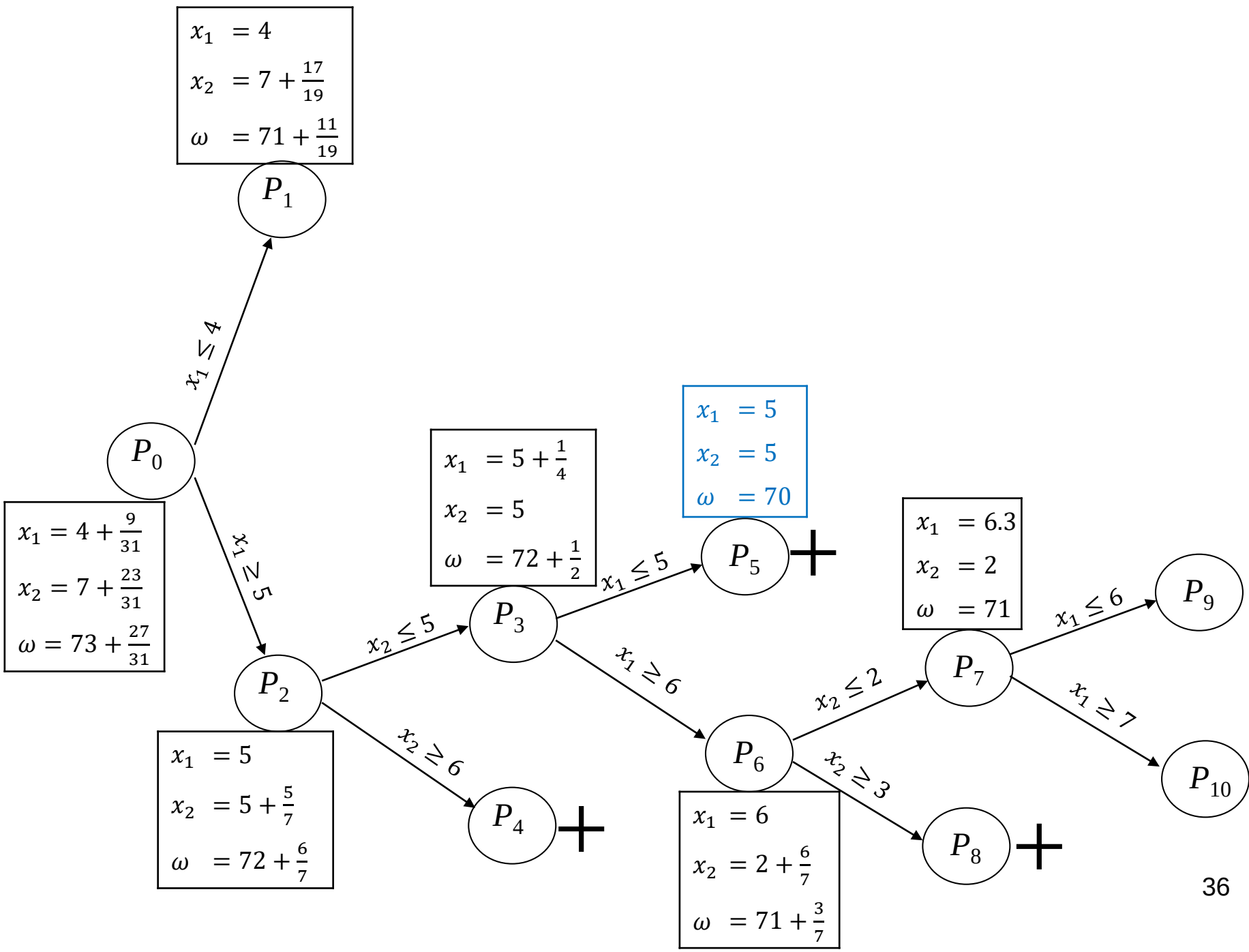


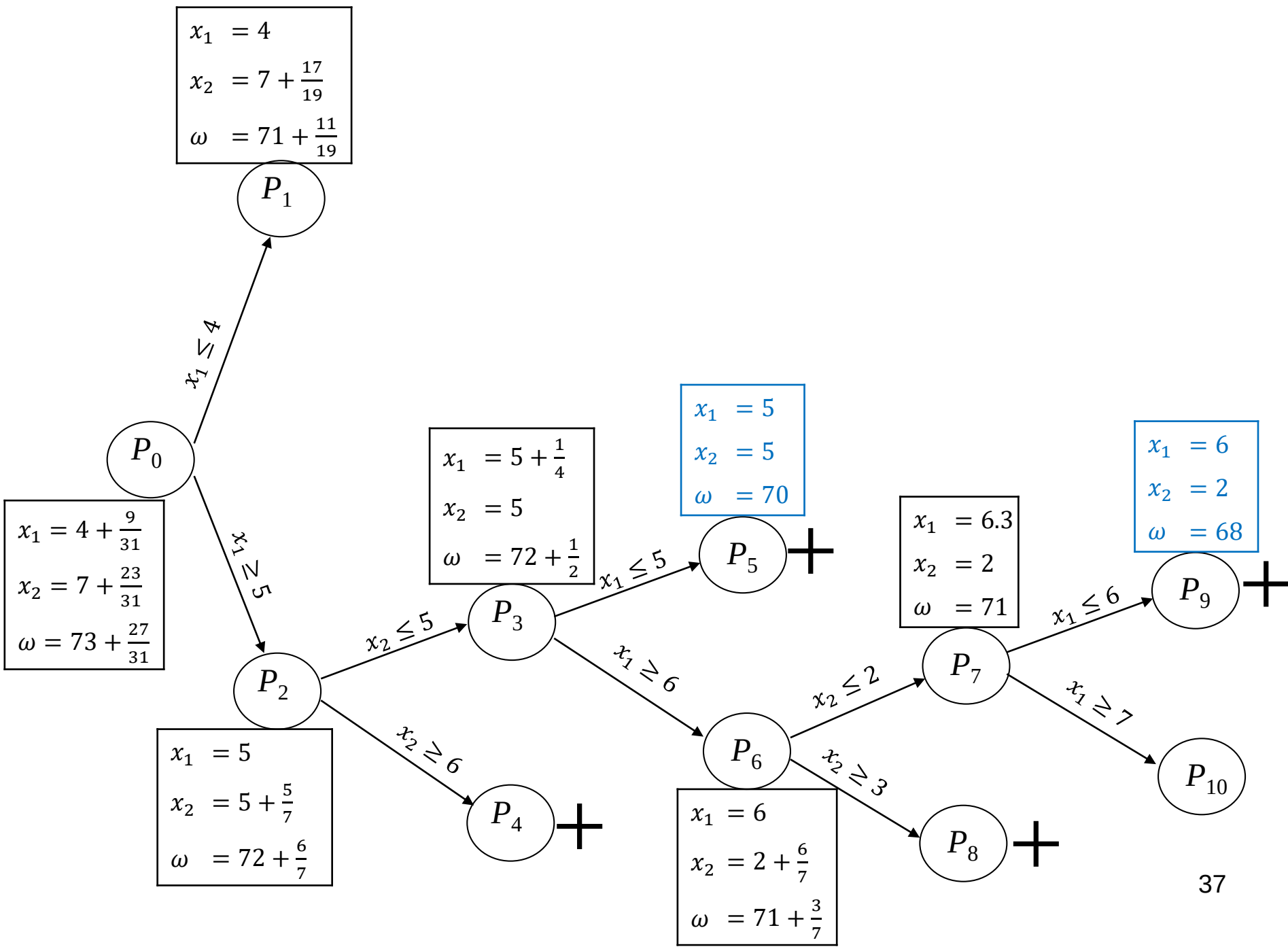


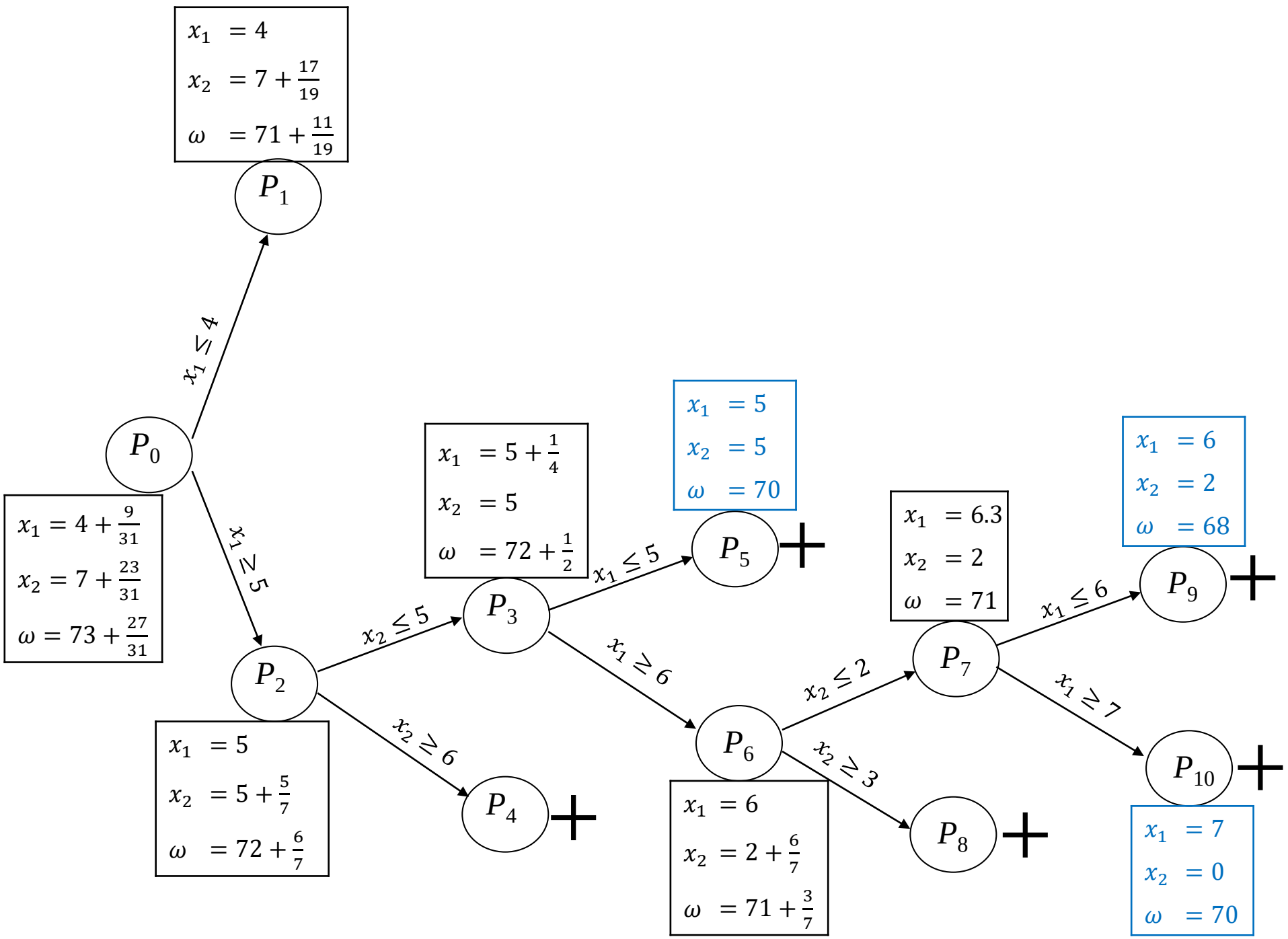


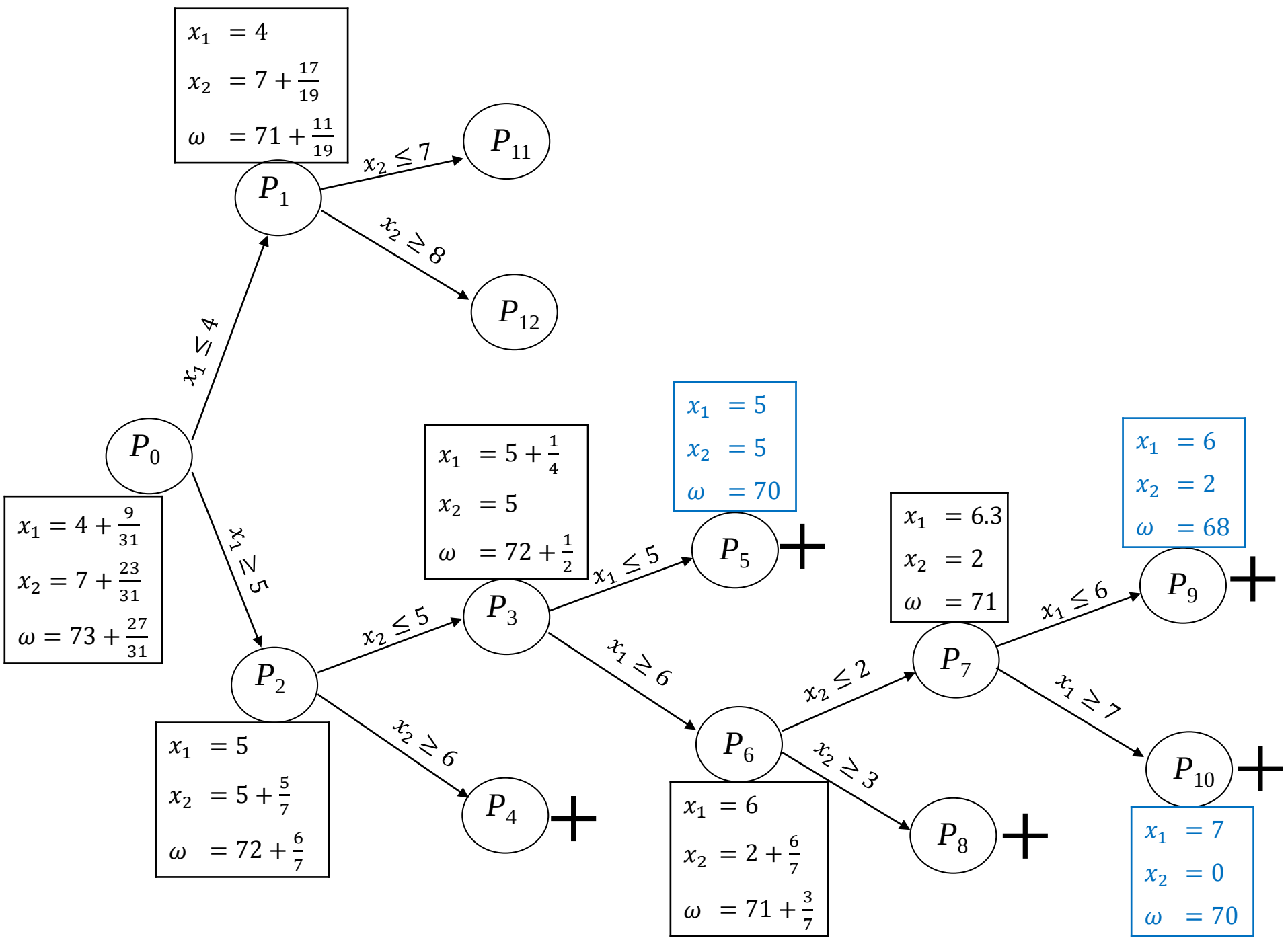


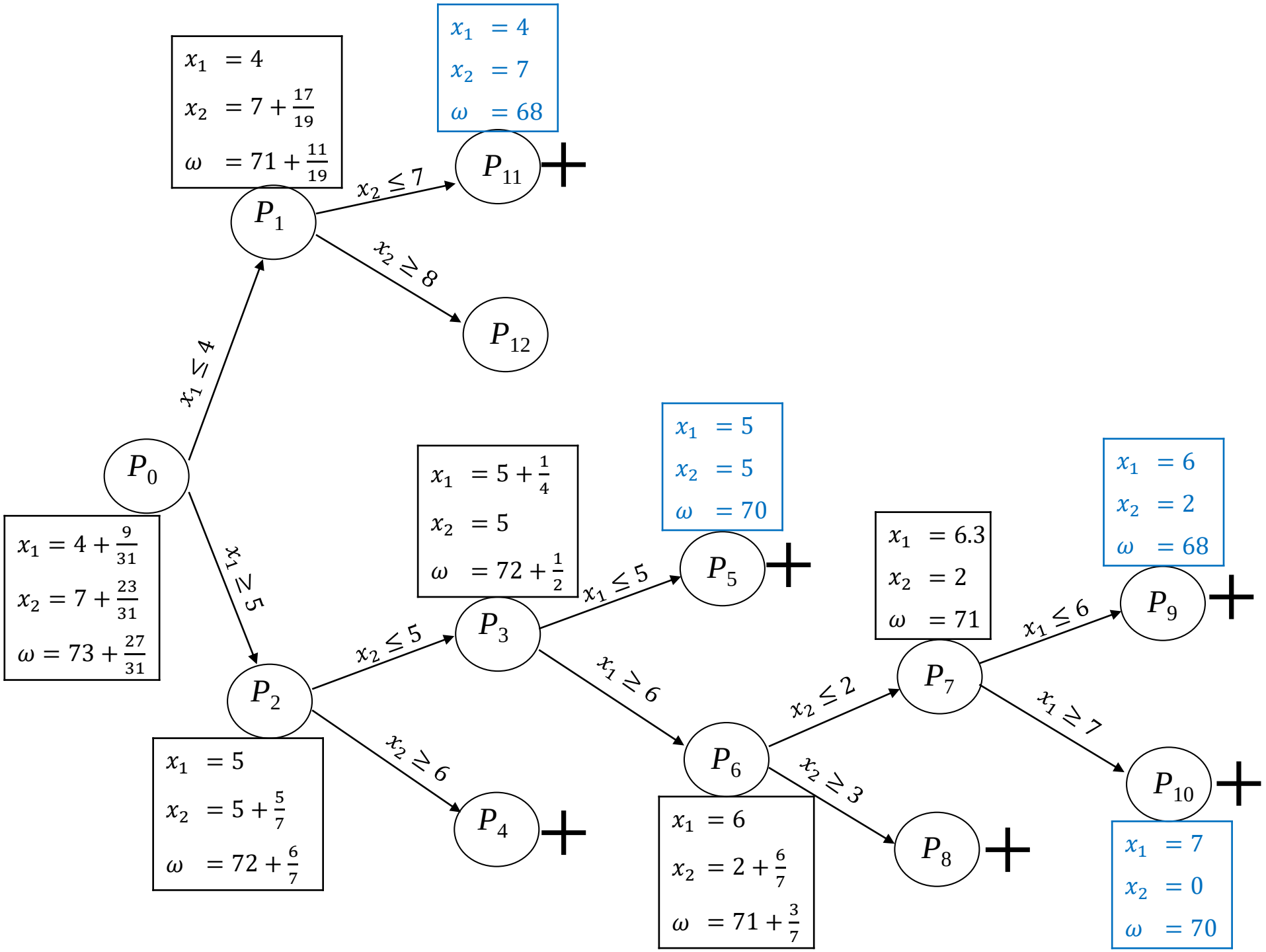


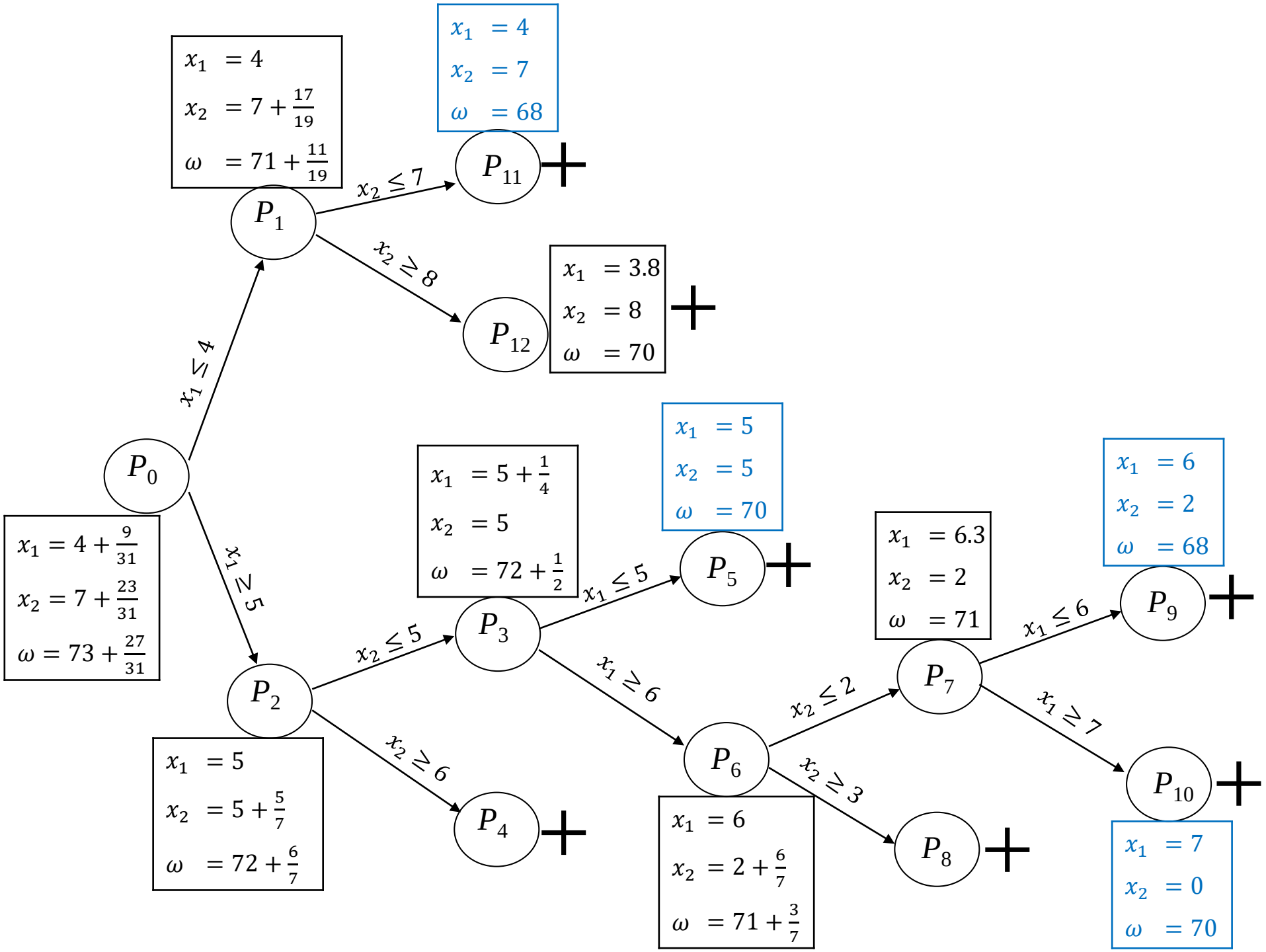












Si consideri il PLI

$$\max \omega = 5 x_1 + 2 x_2$$

$$2 x_1 + 2 x_2 \leq 9$$

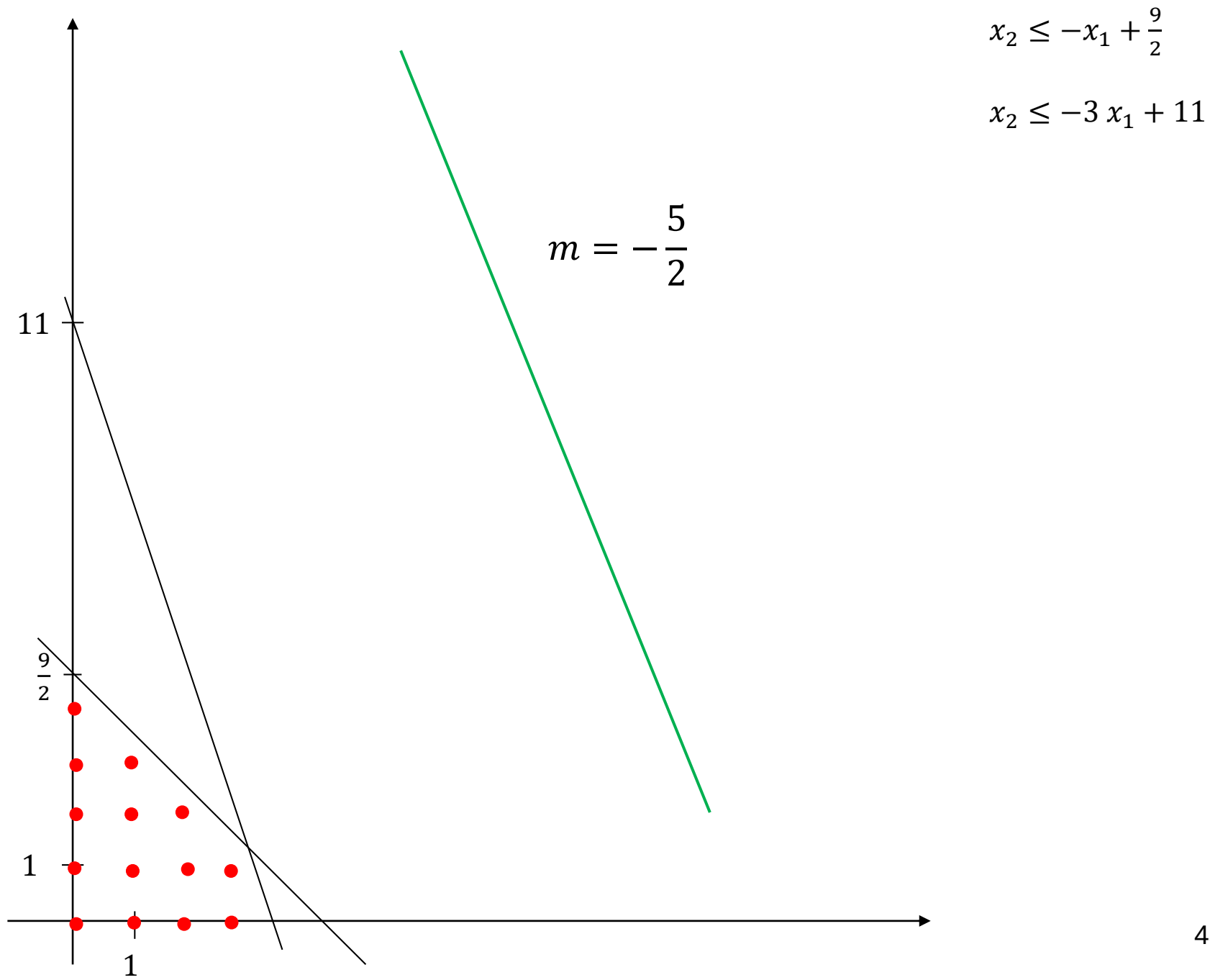
$$3 x_1 + x_2 \leq 11$$

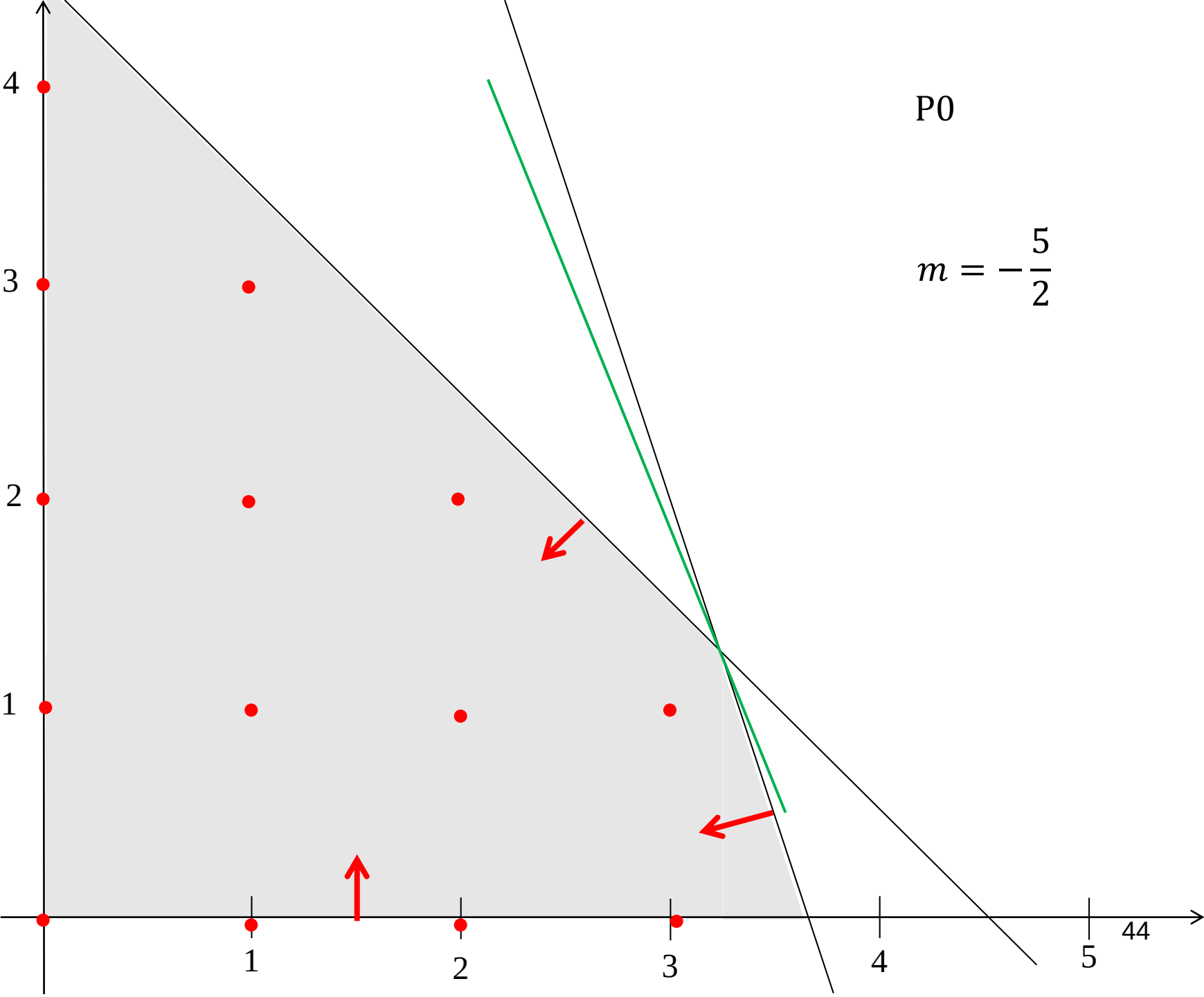
$$x_1, x_2 \geq 0 \text{ e intere}$$

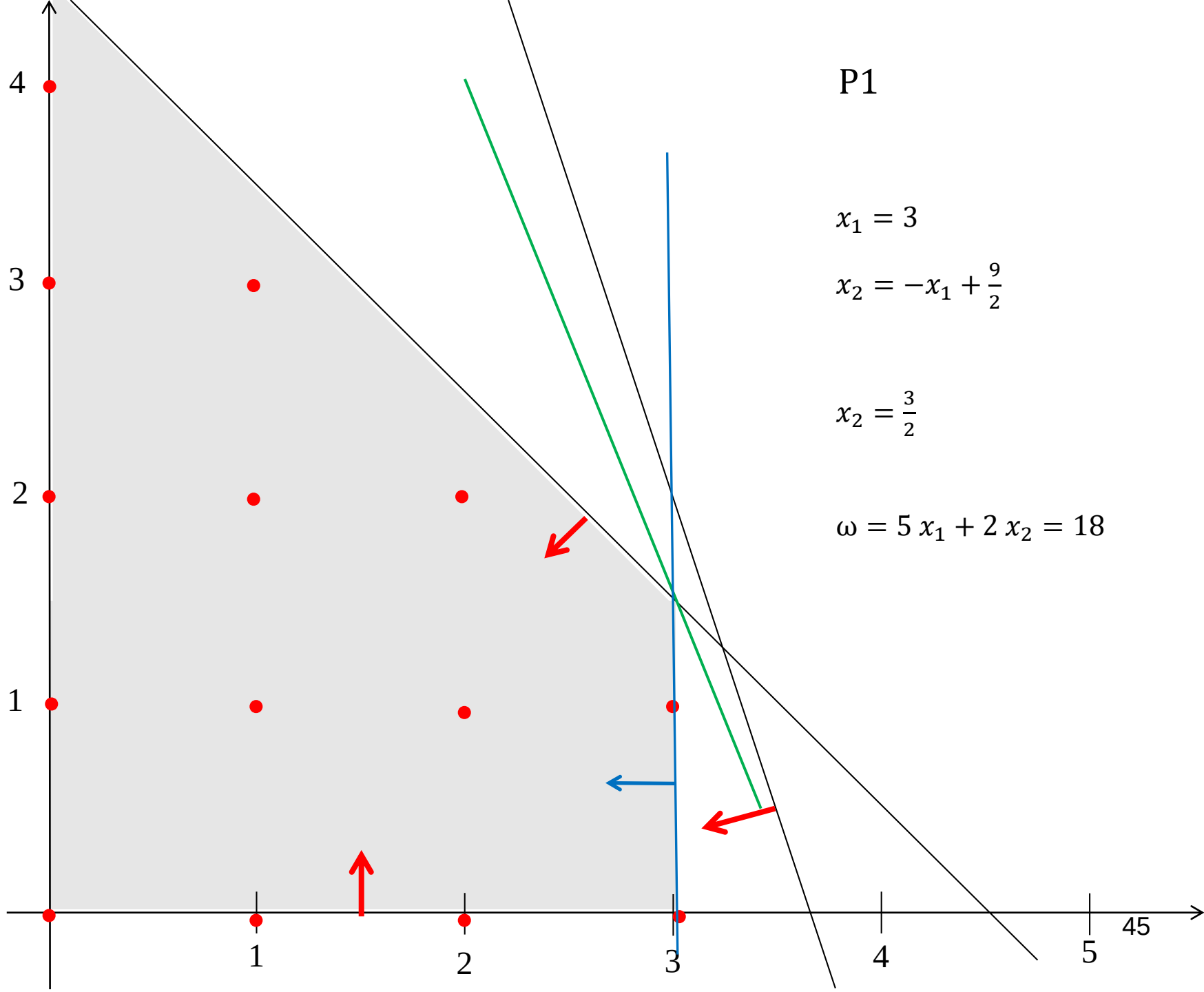
$$x_2 \leq -x_1 + \frac{9}{2}$$

$$x_2 \leq -3 x_1 + 11$$

$$x_2 \leq -\frac{5}{2} x_1 + \frac{\omega}{2}$$







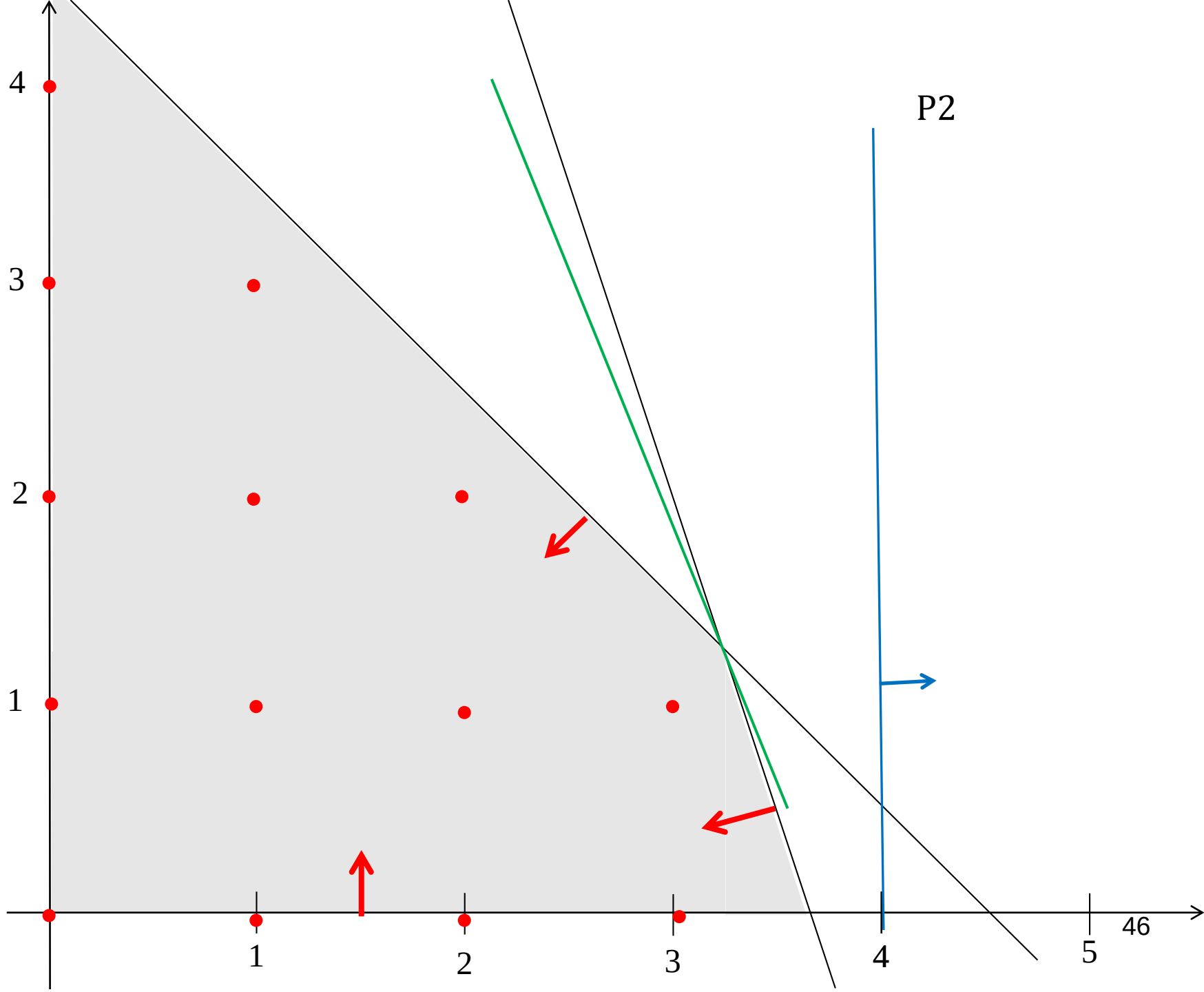
P1

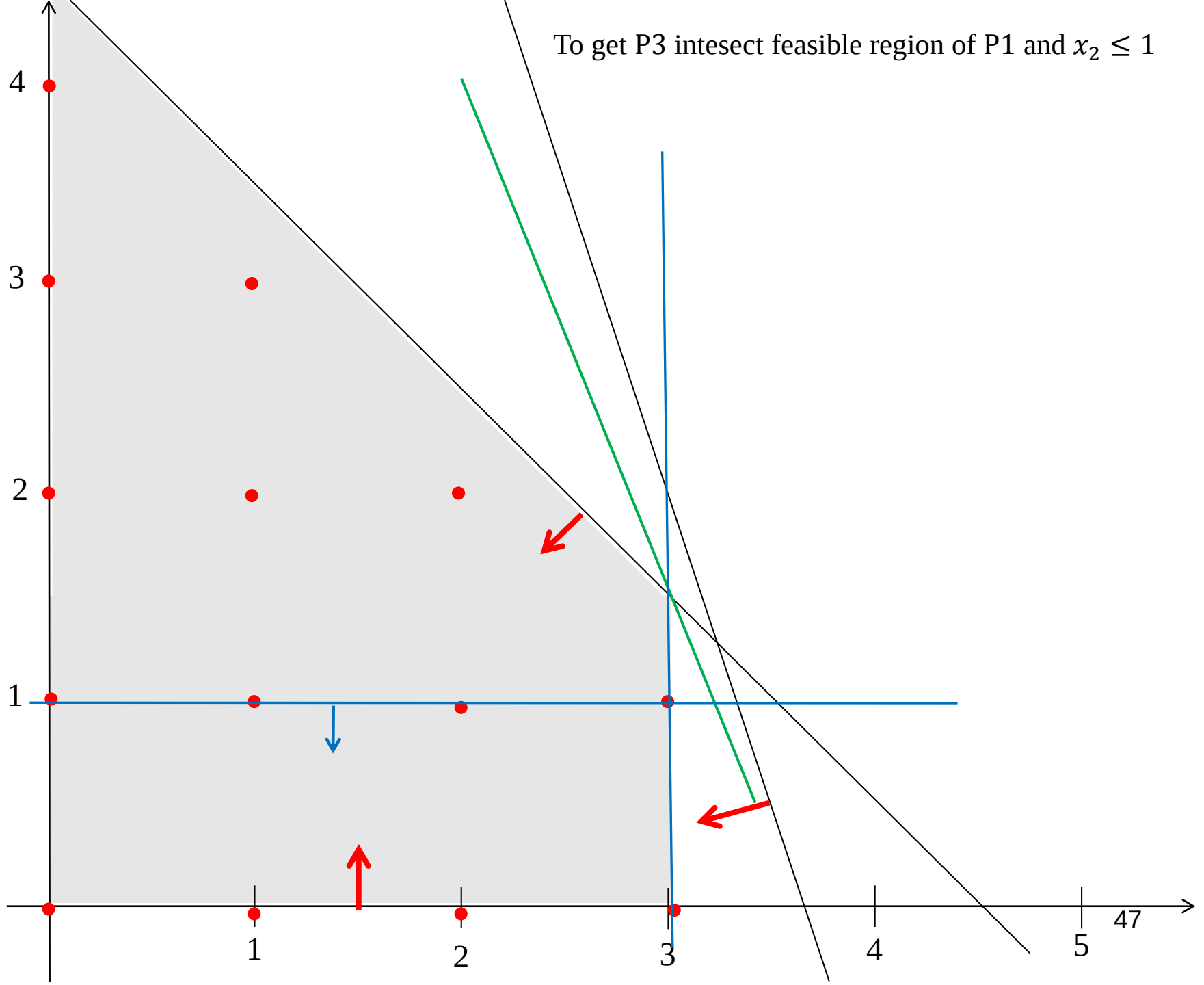
$$x_1 = 3$$

$$x_2 = -x_1 + \frac{9}{2}$$

$$x_2 = \frac{3}{2}$$

$$\omega = 5x_1 + 2x_2 = 18$$



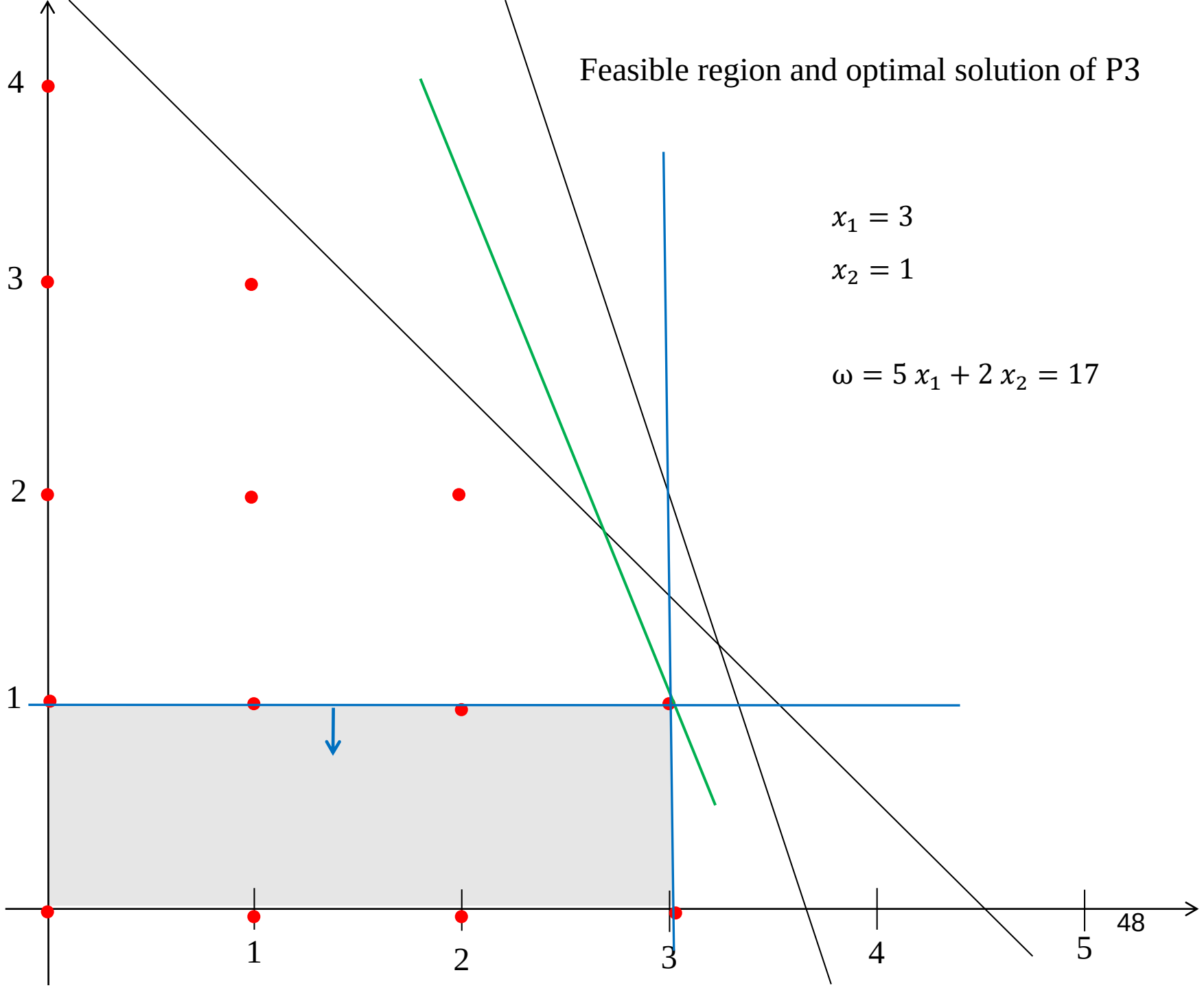


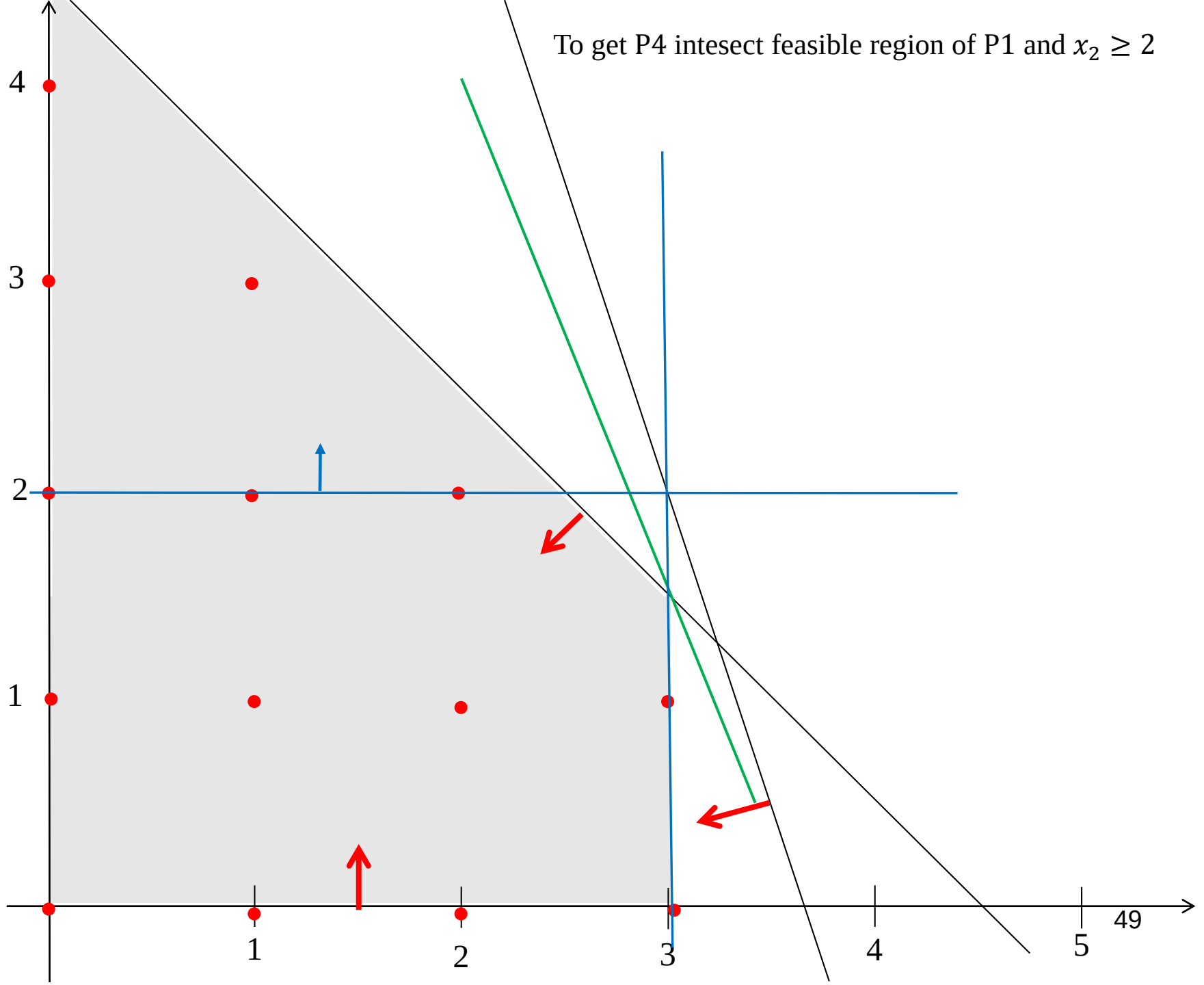
Feasible region and optimal solution of P3

$$x_1 = 3$$

$$x_2 = 1$$

$$\omega = 5x_1 + 2x_2 = 17$$





Feasible region and optimal solution of P4

$$x_2 = 2$$

$$x_2 = -x_1 + \frac{9}{2}$$

$$x_1 = -x_2 + \frac{9}{2} = \frac{5}{2}$$

$$\omega = 5x_1 + 2x_2 = \frac{33}{2} = 16.5$$

